

Holding the invisible: Advanced materials for hydrogen storage

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ABSTRACT

Hydrogen storage remains the main barrier to the broader use of hydrogen as an energy carrier, despite hydrogen's high energy density and clean combustion. This study presents a comparative evaluation of conventional and emerging storage methods, integrating thermodynamic, kinetic, economic, and environmental metrics to assess capacity, efficiency, cost, and reversibility. Physisorption analysis reveals that metal organic frameworks can achieve storage capacities up to 14.0 mmol/g. Chemical storage systems are evaluated, including nanostructured MgH_2 (7.6 wt%), catalyzed reversible complex hydrides, liquid organic hydrogen carriers, and clathrate hydrates. Techno-economic analysis shows storage costs from \$500–700/kg H_2 to \$30–50/kg H_2 , with energy efficiencies of 50%–90%. Life cycle assessment identifies manufacturing as the primary source of emissions, with carbon footprints varying from 150 to 2057 kg CO_2 -eq/kg H_2 . Cryo-compressed is the most practical transportation option while metal hydrides suit stationary use. This study provides a quantitative foundation to guide material selection and system design for next-generation hydrogen storage technologies.

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1. Introduction

The global transition toward cleaner and more sustainable energy sources has elevated hydrogen as a promising energy carrier. Its unique characteristics, including abundance, high energy content per unit mass, and environmentally benign combustion (producing only water vapor), make it a valuable energy resource [1]. Hydrogen is anticipated to play a crucial role across multiple sectors, including transportation, energy storage, and industrial processes over the next few decades [2]. Recent data from the International Energy Agency (IEA) highlights hydrogen's expanding role in the energy sector, as seen in Fig. 1 [3]. In 2023, global hydrogen demand reached a record 97 Mt, representing a 2.5% increase from 2022, driven primarily by refining and chemical industries. This demonstrates the potential for scale-up to 49 Mt by 2030, representing a 30% increase from previous projections. Electrolysis is emerging as a critical technology, with global electrolyzer capacity expected to reach 520 GW by 2030 [3]. However, the practical implementation of hydrogen technologies faces significant challenges, particularly in hydrogen storage. Storage systems must simultaneously meet rigorous criteria, including high gravimetric and volumetric capacities, cost-effectiveness, rapid kinetic performance, and operational safety [4,5]. The low energy density of hydrogen at standard conditions necessitates innovative storage solutions to enable widespread adoption. Storage strategies range from small-scale options (e.g., pressurized tanks for vehicles) to large-scale underground hydrogen storage in geological formations that can buffer seasonal supply-demand variations [6,7]. Underground storage in salt caverns or depleted gas reservoirs provides massive capacity and capitalizes on existing subsurface infrastructure, making it a key component for bulk storage needs in a future hydrogen economy.

Recognizing these challenges, the U.S. Department of Energy (DOE) has established progressive targets for hydrogen storage systems to guide research and development efforts. These targets, detailed in Table 1, demonstrate a strategic approach to improving hydrogen storage technologies across multiple performance parameters, including gravimetric capacity, hydrogen content, volumetric capacity, and system operational characteristics.

This paper provides a comprehensive review of hydrogen storage technologies, systematically examining physical storage methods (including adsorption on porous materials) versus chemical storage methods, which involves hydrogen bonding with other materials and released on-demand. Section 2 discusses physical hydrogen storage, including compressed gas, cryogenic liquid, cryo-compressed hydrogen, and novel physisorption techniques using porous materials. Section 3 covers chemical storage, such as metal hydrides (and their advanced nanostructured variants), complex hydrides, and Liquid Organic Hydrogen Carriers LOHCs (with an emphasis on formic acid). Section 5 provides an in-depth summary of the discussed methods and Sections 6 and 7 dive into techno-economic considerations and life cycle assessments of the various methods. Finally the conclusion identifies research gaps and the future outlook.

2. Physical hydrogen storage

Physical hydrogen storage refers to storing hydrogen in its molecular form without chemical transformation. This includes storage as a compressed gas under high pressure, a cryogenic liquid at extremely

low temperatures, or adsorbed onto high surface area materials such as metal-organic frameworks (MOFs) or activated carbon.

2.1. Compressed hydrogen storage

Compressed hydrogen storage involves storing hydrogen gas at high pressures, typically between 350 and 700 bar, in tanks constructed from advanced materials such as metals or composite polymers to ensure structural integrity under stress [2]. This method is technologically mature and widely deployed due to its operational simplicity and rapid refueling capabilities, which are critical for automotive applications. Modern storage systems use tank designs categorized as Type I (all-metal), Type II (metal liner with composite hoop wrapping), Type III (metal liner fully wrapped with composite), and Type IV (polymer liner fully wrapped with composite). Type IV tanks are lightweight and minimize hydrogen permeation, though they require careful material selection to ensure durability over repeated pressure cycles [8].

Despite their advantages, compressed systems face challenges, including significant energy consumption during compression (10%–15% of the hydrogen's energy content) and high costs linked to carbon fiber composites [5]. Additionally, hydrogen's low volumetric energy density limits practicality in space-constrained applications like light-duty vehicles.

Safety concerns, such as risks of leaks or tank ruptures, necessitate robust engineering solutions, including advanced valve systems and compliance with rigorous safety standards [9]. Recent advancements focus on improving composite materials to enhance gravimetric capacity and reduce environmental impacts. For instance, novel carbon fiber composites and polymer liners have improved impermeability and resistance to hydrogen embrittlement [8]. Hybrid approaches, such as cryo-compressed storage (combining moderate compression with sub-ambient subambient temperatures) and systems integrating hydrogen with other energy carriers like compressed air, aim to optimize efficiency and volumetric capacity [10,11]. Ongoing research, guided by DOE targets, emphasizes innovations in additive manufacturing, advanced resins, and pressure-regulation technologies to reduce costs and improve performance. However, achieving cost-competitive commercial deployment will require further breakthroughs in material science and infrastructure development.

Advancements in materials science are focused on developing stronger, lighter materials for tank construction, such as advanced fiber-reinforced composites, to improve gravimetric capacities and reduce environmental impacts. Additionally, research into alternative tank designs and conformable storage systems aims to optimize space utilization in vehicles [12]. Contemporary studies are also exploring hybrid storage concepts, for instance, combining compressed hydrogen with other energy storage media. Notably, Bartela (2020) [10] proposed a hybrid energy storage system using compressed air and hydrogen to reduce energy consumption and improve efficiency. Nevertheless, the main drawbacks remain the high material costs and energy consumption for compression, along with the relatively low energy density of hydrogen compared to conventional fuels [13,14]. Ongoing research seeks to address these issues through innovative composite materials and advanced pressure regulation systems, including the development of different tank types (Type I–IV) with improved performance characteristics [14].



Fig. 1. Summary of global low-emissions hydrogen activity from 2021 onward. Panel (a) chart tracks production from renewables and fossil fuels with CCUS, showing a 50 percent increase since 2021 and panel (a') highlights announced low-emissions hydrogen projects to 2030, including those at final investment decision. In panel (b), electrolyzer capacity surges from 0.6 GW to over 5 GW by 2024 (a ninefold jump), while announced electrolyzer projects (b') also expand dramatically. Panel (c) shows annual electrolyzer manufacturing capacity quadrupling from 9 GW/yr to 41 GW/yr by 2024, and the charts of (c') details total announced manufacturing capacity by 2030. Finally, panel (d) depicts rising numbers of published hydrogen strategies and (d') shows electrolyzer installation investment reaching an estimated 7.0 billion USD by 2024. [3].

2.2. Liquid (cryogenic) hydrogen storage

Liquid hydrogen storage requires cooling hydrogen gas to cryogenic temperatures below -253°C , converting it into a liquid with significantly higher energy density than its compressed gas form [15,16]. This method enables higher hydrogen storage density, making it ideal for space-constrained applications such as in aerospace [17]. However, liquefaction is highly energy-intensive, consuming about 30%–40% of the hydrogen's energy content due to extreme cooling via Joule–Thomson

expansion and cryogenic refrigeration cycles [2,18]. Boil-off losses arise from gradual hydrogen evaporation due to heat ingress, even with advanced insulation materials like multilayer vacuum insulation, leading to hydrogen loss and potential safety risks [19]. Safety and material challenges also arise because cryogenic temperatures require specialized materials and handling procedures to prevent embrittlement, thermal contraction, and material degradation [20]. The environmental impacts of producing and disposing of cryogenic equipment highlight

Table 1

U.S. department of energy revised targets for onboard hydrogen storage systems.

Source: Adapted from energy.gov.

Storage parameter	Target (2020)	Target (2025)	Ultimate target
Gravimetric capacity (kWh/kg)	1.5	1.8	2.2
Hydrogen content (wt%)	4.5%	5.5%	6.5%
Volumetric capacity (kWh/L)	1.0	1.3	1.7
Hydrogen density (g H ₂ /L system)	30	40	50
Operating temperature (°C)	−40 to +60	−40 to +60	−40 to +60
Min/Max delivery temperature (°C)	−40 to +85	−40 to +85	−40 to +85
Cycle life (cycles)	1,500	1,500	1,500
Fill time for 5 kg H ₂ (min)	3–5	3–5	3–5

the need for more efficient liquefaction processes, better insulation materials, and improved cryogenic handling technologies [16,21].

Innovations in liquefaction technologies focus on reducing energy consumption through advanced refrigeration cycles, utilization of renewable energy sources, and recovery of expansion work [22]. Additionally, research into new insulation materials and tank designs aims to minimize boil-off losses and improve safety [17]. Active cooling systems or cryocoolers are being integrated into storage tanks to reliquefy boil-off gas and prevent losses. While cryogenic hydrogen storage is well established in space applications, it is now under investigation for automotive and stationary energy systems [23]. The key advantage of cryogenic storage is its higher volumetric density, making it suitable for applications requiring high energy density in a compact space [13]. However, as noted, cryogenic storage's energy penalty and boil-off issues limit its standalone viability for everyday use.

2.3. Cryo-compressed hydrogen storage

Cryo-compressed hydrogen storage is a hybrid approach that merges cryogenic cooling with high-pressure compression, offering a middle ground between liquid and high-pressure gas storage. Unlike liquid hydrogen stored near atmospheric pressure or gaseous hydrogen at ambient temperature and high pressure, this method utilizes subcooled hydrogen at elevated pressures to enhance storage efficiency [24]. By partially lowering the temperature, storage pressure can be reduced while maintaining a higher density than conventional compressed gas storage [25]. This approach mitigates boil-off losses associated with liquid hydrogen while improving volumetric energy density beyond that of high-pressure gas storage [26]. In cryo-compressed systems, hydrogen is typically cooled to temperatures anywhere between 20 K and 80 K and then compressed to pressures that can reach around 20–35 MPa [27]. The combination of reduced temperature and moderate pressure substantially elevates hydrogen density in the storage vessel. The storage tank in a cryo-compressed system is usually a vacuum-insulated vessel (similar to a liquid hydrogen Dewar) but also reinforced to withstand significant internal pressure [26]. One can broadly identify variations in design for cryo-compressed tanks: (i) single-walled vessels with very thick insulation designed to minimize heat transfer under moderate pressures, and (ii) multi-layer, composite tanks using advanced carbon-fiber composites to handle the combined stresses of low temperatures and pressurization [25].

Cryo-compressed storage offers longer dormancy, the duration before venting becomes necessary, and achieves higher overall hydrogen densities than either cryogenic or compressed storage alone. Dormancy times can reach weeks under optimized conditions, and hydrogen densities exceeding 40 g H₂/L have been reported, surpassing conventional 700 bar tanks. [26]. This hybrid approach also lowers the effective boil-off: because the tank can tolerate some pressure increase, evaporating hydrogen can be kept in the tank until it is re-cooled or used, rather than vented. Recent research has explored strategies to enhance cryo-compressed system performance. For instance, incorporating a throttling valve for hydrogen venting with cooling recovery can reduce pressure swings and reclaim cooling power from vented gas [28]. Similarly, Aceves et al. 2012 designed cryogenic pressure vessels that

enable safe, long-range vehicle operation, showing that such vessels can be fast refueled and offer improved efficiency compared to conventional storage. Despite these advancements, cryo-compressed hydrogen storage still encounters technical challenges. Highly effective insulation is critical to minimize heat ingress and prevent undesirable pressure rises within the tank. Moreover, integrating both a refrigeration system and high-pressure components introduces design complexity that affects system reliability and cost. Currently, infrastructure capable of delivering subcooled, pressurized hydrogen for vehicle refueling remains underdeveloped. However, prototype vehicles, such as those demonstrated by BMW and Lawrence Livermore National Laboratory, have validated the technical feasibility of cryo-compressed systems. Continued research and development in this area may enable cryo-compressed hydrogen storage to achieve DOE's long-term performance and cost targets [29].

2.4. Refrigeration system optimization for mixed cryogenic compression

As mentioned earlier, cryogenic compressed hydrogen storage combines the advantages of compressed gas and liquid hydrogen storage by maintaining hydrogen in a supercritical state at cryogenic temperatures (70–80 K) and moderate pressures (30–50 MPa). This approach achieves storage densities comparable to liquid hydrogen while eliminating the need for constant energy input to sustain the liquid phase. Recent optimization studies have demonstrated significant improvements in storage efficiency through refrigeration systems, pressure regulation strategies, and integrated system designs.

The most effective refrigeration optimization approaches combine liquid nitrogen pre-cooling with specialized expansion cycles using optimized working fluids. Xu et al. [30] demonstrated that mixed refrigerant systems coupled with gas expansion cycles achieve specific energy consumption as low as 5.87 kWh kg^{−1} H₂ when using neon as the expansion working fluid, representing nearly 50% of the theoretical minimum of 2.92 kWh kg^{−1} H₂. Comparative analysis shows neon outperforms helium (6.15 kWh kg^{−1} H₂) and hydrogen (6.42 kWh kg^{−1} H₂) in expansion cycles due to superior thermophysical properties and heat transfer characteristics. Liquid nitrogen pre-cooling systems provide effective intermediate cooling before final refrigeration, typically reducing overall energy consumption by 20%–25% compared to systems relying solely on mechanical refrigeration. Song et al. [31] achieved storage densities of 72.53 kg m^{−3} at 70 K and 50 MPa using helium expansion cycles with liquid nitrogen pre-cooling, demonstrating specific energy consumption of 6.2 kWh kg^{−1} H₂ and exergy efficiency of 47.2%. Heat exchanger network optimization using pinch analysis techniques further improves thermal efficiency by 10%–15% through enhanced heat integration and reduced irreversibilities.

Optimal pressure regulation strategies use multi-stage compression with pressure ratios of 3–4 per stage, minimizing compression work while maintaining acceptable discharge temperatures. Variable pressure operation in the 30–50 MPa range enables dynamic optimization based on operating conditions, achieving energy savings of 8%–12% compared to fixed pressure systems. Advanced control systems incorporate real-time thermodynamic property calculations and predictive

algorithms to maintain optimal conditions while ensuring safe operation under varying demand profiles. System integration optimization employs genetic algorithms to simultaneously optimize 25–30 key variables including compression ratios, refrigerant compositions, heat exchanger configurations, and operating temperatures. This systematic approach identifies optimal design parameters and operating strategies that maximize performance while maintaining operational reliability. In addition, integration of thermal energy storage systems and waste heat recovery further improves overall efficiency and provides operational flexibility for varying load conditions.

Large scale comparative studies demonstrate clear performance advantages for optimized refrigeration cycles. Yang et al. [32] compared Claude and Brayton cycles for 120 t/d hydrogen liquefaction systems, showing that Claude cycles achieve 6.6% lower exergy losses (18.98 MW vs. 20.24 MW) and 4.4% lower specific energy consumption (5.62 vs. 5.88 kWh/kg LH₂) while requiring 10.6% less refrigerant flow. The superior performance results from more efficient expansion processes and better integration of compression and expansion systems.

Hybrid system approaches incorporating absorption cooling cycles demonstrate substantial performance improvements. Ghorbani et al. [28] achieved specific energy consumption reduction from 10.78 to 7.948 kWh/kg LH₂ (26.3% improvement) with efficiency increases from 12.05% to 16.34% through integration of absorption cooling with conventional refrigeration systems. These hybrid approaches provide enhanced operational flexibility and improved part-load performance characteristics.

The comprehensive performance comparison of different cryogenic hydrogen storage systems is illustrated in Fig. 2, which demonstrates the relationship between specific energy consumption and exergy efficiency for various optimization approaches. The figure clearly shows that the Claude cycle achieves the highest exergy efficiency (52.8%) while maintaining relatively low energy consumption (5.62 kWh/kg H₂), positioning it closest to the theoretical minimum energy requirement. Mixed refrigerant systems with neon expansion demonstrate the lowest specific energy consumption (5.87 kWh/kg H₂) among expansion cycle configurations, while conventional systems exhibit significantly higher energy requirements (10.78 kWh/kg H₂) with correspondingly lower efficiency (31.2%). The bubble sizes in the figure represent storage densities, with optimized supercritical systems achieving the highest volumetric storage capacities (72.53 kg/m³).

The systematic optimization of mixed cryogenic compressed hydrogen storage systems follows a structured approach that addresses each major system component and their interactions. The first step involves comprehensive system modeling and simulation using advanced process simulation software such as Aspen HYSYS or CHEMCAD to establish baseline performance characteristics and identify optimization opportunities. The modeling process must accurately represent the thermodynamic properties of hydrogen at cryogenic conditions, including real gas effects, phase behavior, and transport properties.

The second optimization step focuses on component-level optimization to maximize the performance of individual system elements before addressing system-level interactions. Compressor optimization involves the selection of optimal compression ratios, intercooling strategies, and capacity control methods to minimize energy consumption while maintaining required pressure levels. Heat exchanger optimization addresses heat transfer enhancement, pressure drop minimization, and thermal effectiveness maximization through advanced design techniques and materials selection. Refrigeration system optimization evaluates different cycle configurations, working fluid selections, and operating conditions to achieve maximum cooling capacity with minimum energy input.

The third step implements system-level optimization that considers the interactions and trade-offs between different components and subsystems. Multi-objective optimization algorithms, such as genetic algorithms or particle swarm optimization, systematically evaluate thousands of potential system configurations to identify optimal design

and operating parameters. The optimization process typically considers 25–30 key variables simultaneously, including compression ratios, refrigerant compositions, heat exchanger configurations, and operating temperatures and pressures.

The fourth step involves dynamic optimization that addresses system performance under varying operating conditions and transient scenarios. Dynamic optimization considers factors such as startup and shutdown procedures, load-following capabilities, and response to disturbances or equipment failures. The development of optimal control strategies ensures that the system maintains peak performance across all operating modes while providing adequate safety margins and operational flexibility.

The final optimization step focuses on economic optimization that balances performance improvements with capital and operating cost considerations. Life cycle cost analysis evaluates the total cost of ownership including capital costs, operating expenses, maintenance requirements, and end-of-life considerations. The economic optimization process identifies the optimal balance between performance and cost, ensuring that optimization efforts focus on improvements that provide the greatest economic benefit. The most advanced optimized systems achieve storage densities of 72.53 kg m⁻³ with specific energy consumption approaching 5.87 kWh kg⁻¹ H₂ and exergy efficiencies exceeding 49%, demonstrating substantial improvements over conventional storage methods and the potential for mixed cryogenic systems to provide an optimal balance between storage density and energy efficiency for large-scale hydrogen storage applications.

2.5. Physical adsorption in porous materials

Physisorption-based hydrogen storage relies on the adsorption of hydrogen onto porous materials at cryogenic temperatures (~ 77 K). This approach stores hydrogen at relatively low pressures by exploiting the large surface areas of adsorbent materials [33]. Commonly studied adsorbents include carbon-based nanostructures, metal–organic frameworks (MOFs), covalent organic frameworks (COFs), and novel host materials like clathrate hydrates. Although inherently safer due to their low-pressure operation and modular design, these systems require cryogenic cooling because hydrogen exhibits weak adsorption energies on these surfaces, limiting effective storage at higher temperatures.

2.5.1. Carbon-based materials

Carbon materials, including activated carbons, carbon nanotubes (CNTs), fullerenes, and graphene, have been extensively investigated for hydrogen storage due to their high surface areas, tunable pore structures, and chemical stability [34–36]. Early studies reported exceptionally high hydrogen uptake in certain carbon nanostructures, for instance, Dillon et al. (1997) [37] claimed hydrogen storage capacity of approximately 5–7 wt% in CNTs at ambient conditions. However, subsequent research revealed that achievable adsorption capacities are closely linked to specific surface area and pore size distribution, and that initial high values were not reproducible [38,39]. Typically, there is about 1 wt% hydrogen storage per 500 m²/g of surface area at 77 K, known as the Chahine rule. This implies that ultra-high surface area carbons (3000 m²/g or more) are necessary to approach the DOE targets via physisorption, achievable only at cryogenic temperatures.

Enhancing hydrogen uptake in carbon materials has been focused on structural tailoring, including templating for ordered porosity, chemical activation to increase micropore volume, and heteroatom or metal doping to strengthen hydrogen interactions [40–42]. For example, activated carbon derived from biomass waste can achieve surface areas above 3000 m²/g and show hydrogen capacities ~ 6 wt% at 77 K and 40 bar [43]. However, practical applications of carbon adsorbents are limited by the need for cryogenic temperatures (typically liquid nitrogen temperature, 77 K) to achieve significant hydrogen uptake, and by challenges in producing materials with consistently high performance.

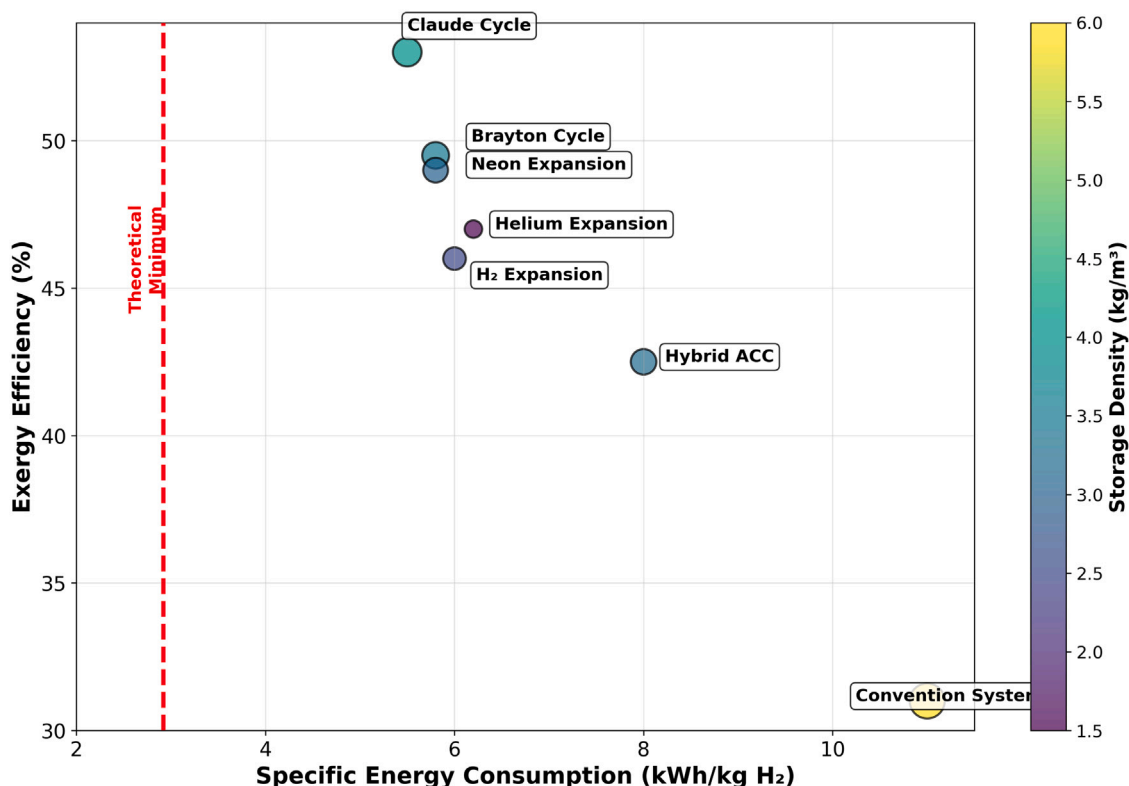


Fig. 2. Performance comparison of cryogenic hydrogen storage systems showing the relationship between specific energy consumption and exergy efficiency. Bubble sizes represent storage densities (kg/m^3). The red dashed line indicates the theoretical minimum energy consumption (2.92 kWh/kg H_2). Data points represent mixed refrigerant systems with different expansion working fluids (neon, helium, hydrogen), large-scale refrigeration cycles (Claude, Brayton), hybrid absorption cooling systems (ACC), and conventional storage systems.

At ambient temperature, the uptake on carbon is only a few tenths of a percent at best, which is far below requirements [44].

Recent advancements include the development of single-walled carbon nanohorns (SWNHs), which are similar to single-walled carbon nanotubes but consist of horn-shaped structures. SWNHs have been investigated for hydrogen storage due to their high purity and unique porous structures [45]. Studies have shown that SWNHs can adsorb hydrogen through physical adsorption mechanisms, with adsorption capacities influenced by their porosity and surface area. Additionally, research into carbide-derived carbons (CDCs) has demonstrated that materials with subnanometer pores can store up to 3 wt% hydrogen at 60 bar and -196°C , with further increases possible through chemical or physical activation [46]. While carbon-based materials offer promising avenues for hydrogen storage due to their structural versatility and high surface areas, achieving practical storage capacities at ambient conditions remains a significant challenge. Ongoing research is focused on optimizing pore structures, surface chemistries, and material compositions to enhance hydrogen adsorption properties, with the goal of meeting the stringent requirements for practical hydrogen storage applications.

2.5.2. Metal-organic frameworks (MOFs)

MOFs are crystalline materials composed of metal ions coordinated to organic ligands, forming porous structures with exceptionally high surface areas (often $3,000\text{--}6,000 \text{ m}^2/\text{g}$) [47,48]. Their tunable pore sizes, high porosity, and ability to incorporate functional sites make them attractive for hydrogen storage [49,50]. MOFs like MOF-5, MOF-177, and IRMOF-20 have demonstrated promising hydrogen storage capacities under cryogenic conditions. Strategies to enhance hydrogen storage in MOFs include increasing surface area through elongated linkers, incorporating unsaturated open metal sites to increase hydrogen binding energies, and functionalizing pore surfaces with polar

groups or metal additives to create stronger adsorption sites [51]. For example, introducing open metal sites (e.g., in HKUST-1 or MOF-74) can increase the isosteric heat of adsorption and improve hydrogen uptake, especially at higher temperatures than 77 K [52].

Despite their potential, effective hydrogen storage in MOFs still typically requires cryogenic temperatures. The physisorption binding energies in MOFs (generally $\sim 4\text{--}8 \text{ kJ/mol}$ for H_2) are insufficient for significant adsorption at ambient temperature, at 298 K even the best MOFs adsorb only a small fraction of their 77 K capacity [53]. Additionally, practical implementation of MOFs is hindered by issues such as limited material stability, many MOFs are sensitive to moisture or heat as well as challenges in large-scale synthesis and the high cost of precursors. Recent advances, however, show promising progress; a pore-engineered MOF was reported to achieve a record hydrogen excess uptake exceeding $7.0 \text{ wt\%}$ at 77 K and 100 bar [54], underscoring ongoing development toward viable sorbent-based storage systems.

2.5.3. Covalent organic frameworks (COFs) and polymers

COFs and porous organic polymers (such as Conjugated Microporous Polymers (CMPs), Hyper-Crosslinked Polymers (HCPs), Porous Aromatic Frameworks (PAFs), etc.) are constructed from light elements (H, B, C, N, O) linked by strong covalent bonds, resulting in materials with high surface areas and low densities [55]. They provide tunable pore sizes, structural diversity, and lightweight nature, making them promising for hydrogen storage applications [56,57]. Similar to MOFs, strategies to enhance hydrogen storage in COFs involve increasing surface area, optimizing pore size (to maximize overlap of adsorption potentials), and incorporating functional groups or dopants to enhance hydrogen interactions [58,59]. For example, porous aromatic frameworks doped with exposed metal sites have been studied to increase binding energies. However, as with other physisorbents, the hydrogen binding in COFs is weak; significant storage capacity is only observed at

Table 2

Temperature-dependent hydrogen storage performance of representative MOFs and COFs.

Material	77 K (wt%)	195 K (wt%)	233 K (wt%)	273 K (wt%)	298 K (wt%)
MOF-5	7.1	2.8	1.9	1.2	0.8
MOF-177	11.2	4.1	2.7	1.6	1.0
HKUST-1	6.9	3.2	2.4	1.8	1.3
MOF-74(Mn)	8.6	4.8	3.9	3.1	2.4
NU-1501	14.0	5.2	3.4	2.1	1.3
COF-102	8.1	2.9	1.8	1.0	0.6
COF-103	9.7	3.4	2.2	1.3	0.8
PAF-1	10.3	3.7	2.4	1.4	0.9

cryogenic temperatures. Material stability under operating conditions (many COFs are powders that may pack poorly or have flexibility that leads to pore collapse) and challenges in large-scale synthesis remain obstacles [60]. Research is ongoing to create COFs with more robust architectures and to infiltrate COF pores with tiny amounts of catalyst or metal to enhance hydrogen uptake kinetics and reversibility.

2.5.4. Adsorption energy and temperature effects in porous materials

Optimizing hydrogen storage in porous materials requires tuning adsorption energies through complementary strategies. The ideal binding range, 15–25 kJ mol⁻¹, balances ambient-temperature retention with efficient desorption, well above typical physisorption energies (4–8 kJ mol⁻¹) in pristine materials. Open metal sites offer a direct route to boosting adsorption energies. Varying metal centers in isostructural MOFs like M₂(dobdc) (MOF-74 analogs) shows clear trends: Mn²⁺ (22.1), Fe²⁺ (19.8), Co²⁺ (17.2), Ni²⁺ (15.1), Cu²⁺ (12.8), and Zn²⁺ (8.9 kJ mol⁻¹), with binding strength tied to metal charge density and electronic structure [61]. This allows precise tuning via metal selection. Matching pore size to the kinetic diameter of H₂ (2.89 Å) improves adsorption by enhancing the overlap of van der Waals potentials from opposing pore walls. Pores in the 6–12 Å range, particularly those around 7–8 Å, exhibit up to 40%–60% higher binding energies. This effect is more pronounced in slit-shaped pores than in cylindrical geometries [62].

Ligand functionalization offers a versatile approach for tuning local binding environments by introducing polar groups, heteroatoms, or pendant metal sites. For example, amine-functionalized MOFs such as mmen-Mg₂(dobpdc) achieve binding energies of 21–25 kJ mol⁻¹ through weak coordination between hydrogen and nitrogen lone pairs. Lithium incorporation via post-synthetic modification can also yield binding energies of 18–22 kJ mol⁻¹ while preserving structural stability and reversibility [63]. Controlling framework interpenetration provides another means of optimizing adsorption. Non-interpenetrated structures tend to have larger pores but weaker interactions, whereas tailored interpenetration can produce ideal pore dimensions with enhanced adsorption potentials. This parameter can be systematically adjusted through ligand design and synthesis conditions [64]. The practical viability of porous hydrogen storage materials depends heavily on their performance from cryogenic (77 K) to ambient (298 K) conditions. Temperature-dependent analysis is essential for identifying performance limits and refining material design. At 77 K, top-performing MOFs approach theoretical storage limits. MOF-210 achieves a total uptake of 17.6 wt% at 80 bar, and NU-1501 reaches 14.0 wt% under similar conditions. These capacities reflect near-complete pore filling, as storage is limited by available pore volume rather than binding strength under cryogenic conditions [65]. (See Table 2.)

The transition from cryogenic to intermediate temperatures (195–233 K) highlights the importance of binding energy in sustaining storage capacity. MOFs with stronger interactions, such as MOF-74(Mn), retain 56% of their 77 K capacity at 195 K, compared to only 39% for MOF-5, which relies on weaker physisorption. This performance gap widens with temperature, underscoring the need for

optimized binding energies in practical systems. At near-ambient conditions (273–298 K), current porous materials show clear limitations. Even top-performing MOFs reach only 2–3 wt% uptake at 298 K and 10–50 bar. MOF-74(Mn) achieves 2.4 wt% at 50 bar, attributed to its strong open metal sites and favorable pore geometry, while MOF-5 stores less than 1 wt% under similar conditions [61]. Covalent organic frameworks (COFs) follow similar temperature trends but generally offer lower capacities due to the absence of metal-based binding sites. COF-102 and COF-103 show promising cryogenic uptake (8.1 and 9.7 wt% at 77 K, respectively) but drop to 0.6–0.8 wt% at 298 K. Their purely organic structure limits binding energies to the physisorption range, requiring new design strategies for ambient-temperature applications [66]. Porous aromatic frameworks (PAFs) serve as a middle ground between MOFs and COFs, combining high surface areas with moderate binding energies via π - π interactions and micropore confinement. PAF-1 achieves 10.3 wt% at 77 K but retains only 0.9 wt% at 298 K, illustrating the limitations of physisorption-based storage. Still, their exceptional chemical and thermal stability makes PAFs appealing for applications where durability outweighs capacity [67].

Temperature-dependent performance analysis reveals key principles for designing porous materials with improved hydrogen storage at ambient conditions. The main challenge is to achieve binding energies in the optimal 15–25 kJ mol⁻¹ range while maintaining high surface areas and favorable pore structures. Multi-site binding strategies offer a promising route to enhance room-temperature uptake. Materials that combine open metal sites with functionalized organic linkers exhibit synergistic binding, with total interaction energies nearing the sum of individual contributions. For instance, amino-functionalized MOF-74 variants reach 20–24 kJ mol⁻¹, achieving 3.2–3.8 wt% at 298 K and 50 bar—a 50–60% improvement over unmodified forms [63]. Hierarchical pore architectures further optimize both high-pressure capacity and ambient-temperature retention. Bimodal distributions combine large mesopores for volume with micropores for strong binding. Such MOFs deliver 12–15 wt% at 77 K and retain 2.5–3.0 wt% at 298 K, representing well-balanced performance [64]. Spillover mechanisms via metal nanoparticle incorporation provide another pathway for enhancing ambient-temperature storage. Platinum or palladium nanoparticles disassociate H₂ and enable transfer to nearby binding sites, increasing local hydrogen concentration and effective binding energy. This strategy has shown 2–4× capacity improvements at room temperature while preserving usable kinetics [68].

2.5.5. Clathrate hydrates

Clathrate hydrates are crystalline, ice-like solids where hydrogen molecules are physically trapped within the cavities formed by a lattice of water molecules [69]. They have the potential for hydrogen storage at near-ambient temperatures and pressures, as the water lattice stabilizes the hydrogen, making them attractive for practical applications if their storage capacity can be improved [70]. Traditional clathrate hydrates of hydrogen, however, have very low hydrogen storage capacities (< 0.5 wt% typically) and require high formation pressures (often 100–200 MPa for pure hydrogen hydrates), along with slow kinetics of formation and release [71]. These factors have historically made them impractical.

Su et al. (2008) [72] developed a method to enhance hydrogen clathrate formation using an ultralow-density emulsion-templated polymer (polyHIPE foam) as a scaffold [72]. The macroporous structure with interconnected open cells provided abundant nucleation sites and short diffusion pathways, significantly improving hydrate formation kinetics. This approach achieved a hydrogen storage capacity of 0.18 wt% at 270 K (just below room temperature), with 90% of that capacity reached within 60 min, far surpassing conventional hydrate formation rates. However, this capacity remains orders of magnitude below the DOE target (> 5 wt%).

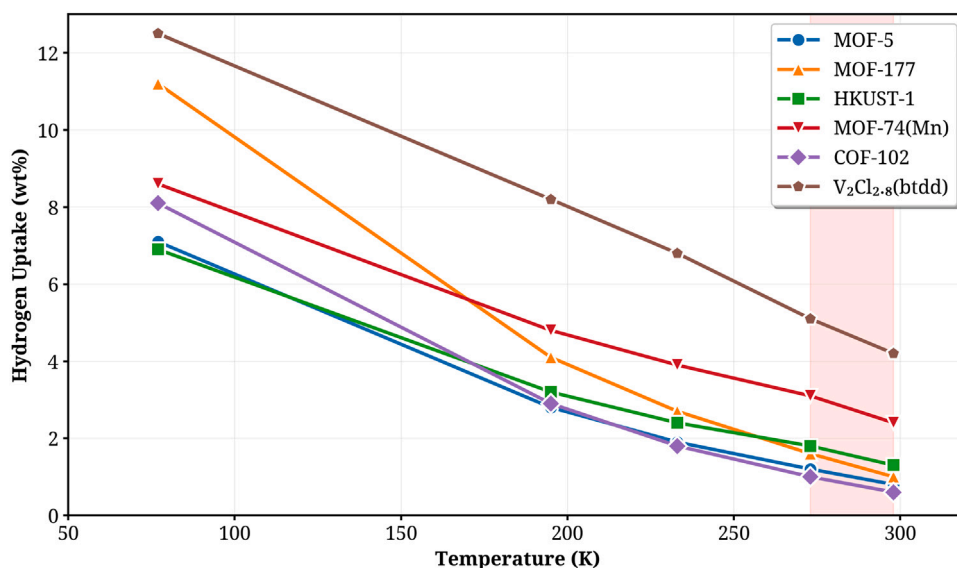


Fig. 3. Temperature-dependent hydrogen storage performance trends for representative MOFs and COFs. Data points represent typical performance ranges reported in literature, with some values estimated based on established trends and binding energy relationships. The dramatic capacity loss from cryogenic (77 K) to ambient (298 K) conditions illustrates the fundamental challenge of physisorption-based storage systems. Materials with enhanced binding energies (e.g., MOF-74 series with open metal sites) demonstrate superior capacity retention compared to purely physisorption-based frameworks. The shaded region indicates the ambient temperature range (273–298 K) where practical applications require operation.

Recent work continues to show both the challenges and the potential breakthroughs in clathrate hydrate research. Notably, Farrando-Perez et al. (2022) demonstrated that using a tailored activated carbon scaffold can dramatically reduce the pressure needed for hydrogen hydrate formation (by up to 30%) and achieve nearly complete water-to-hydrate conversion within minutes, highlighting the power of nanoconfinement to boost formation kinetics [73]. Meanwhile, Nguyen et al. (2021) showed that host-guest hydrogen bonding in clathrates can induce “liquid-like” water phases within an ostensibly crystalline structure—an effect that begins below the typical equilibrium dissociation temperature, suggesting that certain polar guest molecules may partially disrupt the lattice and lower its stability threshold [74]. Comprehensive reviews, such as that by Gupta et al. (2021), also stress the advantages of hydrogen clathrates—particularly their low production cost, reversible nature, and low flammability risk, while acknowledging that further progress is needed to meet the US Department of Energy targets through strategies like promoter additives, nanoconfinement, and hybrid material approaches [75]. Current research thus remains focused on refining clathrate formers, optimizing scaffold design, and exploiting host-guest interactions to increase hydrogen encapsulation and reduce the pressures required for hydrate stability [76]. To date, clathrate hydrates remain an intriguing avenue for hydrogen storage, underscoring the need for continued breakthroughs to realize their full potential.

2.5.6. Comparing mechanisms of physical adsorption

The comparative analysis of hydrogen adsorption mechanisms across carbon materials, MOFs, and COFs reveals fundamental differences in site heterogeneity, binding energy distributions, and optimization strategies [77]. Carbon materials exhibit the most heterogeneous adsorption environments due to their disordered structure and variable surface chemistry, resulting in broad binding energy distributions that can be advantageous for storage applications requiring operation over wide pressure ranges [78]. The micropore filling mechanism in activated carbons provides enhanced binding energies compared to open surfaces and is accurately described by the Dubinin–Astakhov model [79,80]:

$$W = W_0 \exp \left[- \left(\frac{A}{E} \right)^n \right]$$

where W is the volume of adsorbed hydrogen, W_0 is the total micropore volume, A is the adsorption potential, E is the characteristic energy, and n is the heterogeneity parameter typically ranging from 1.5–3 for hydrogen adsorption on carbons [80]. This model effectively captures the micropore filling behavior that dominates hydrogen storage in activated carbons, where the lack of structural control limits optimization possibilities but the heterogeneous pore structure provides storage over wide pressure ranges.

MOFs demonstrate the highest degree of adsorption site diversity, with multiple distinct site types exhibiting significantly different binding energies and geometric requirements [81,82]. This heterogeneity enables MOFs to achieve high storage capacities through efficient utilization of available pore volume but also creates challenges for optimization due to the complex interplay between different site types [82]. The presence of open metal sites provides the strongest binding energies available through physisorption mechanisms (8–15 kJ mol^{−1}), making MOFs particularly attractive for near-ambient temperature storage applications [81]. The multi-site adsorption behavior in MOFs is best described by the Langmuir–Freundlich model for heterogeneous surfaces [82]:

$$q = \sum_{i=1}^n \frac{q_{\max,i} b_i P^{1/n_i}}{1 + b_i P^{1/n_i}}$$

where q is the total adsorbed amount, $q_{\max,i}$ is the maximum capacity for site type i , b_i is the affinity constant, P is pressure, and n_i is the heterogeneity parameter for each site type [82]. This model accounts for the sequential filling of different adsorption sites (open metal sites, oxygen sites, aromatic sites) that characterizes MOF adsorption behavior [81].

COFs occupy an intermediate position between carbon materials and MOFs in terms of site heterogeneity and binding energy distributions [83]. The crystalline structure enables precise control over adsorption environments while maintaining the chemical stability advantages of covalent bonding [83]. The uniform site distributions in COFs facilitate predictive modeling and rational design approaches, although the absence of strong binding sites limits their performance compared to MOFs with open metal sites [84]. For most COFs, the relatively uniform pore structure and homogeneous adsorption sites are well described by

Table 3

Comparative hydrogen adsorption performance and theoretical model effectiveness for representative porous materials across carbon-based, MOF, and COF material classes.

Material class	Representative material	Surface area (m ² /g)	H ₂ uptake (77 K, 1 bar, wt%)	H ₂ uptake (298 K, 100 bar, wt%)	Binding energy (kJ/mol)	Optimal model
Activated Carbon	Maxsorb III	3000	2.5	0.8	5.2	Dubinin–Astakhov
Carbon Nanotubes	SWCNT (1.4 nm)	1200	1.8	0.4	6.8	BET + Micropore Filling
Graphene	Functionalized Graphene	2200	1.2	0.2	4.5	Langmuir–Freundlich
MOF	MOF-5	1270	1.3	0.9	4.8	Multi-site Langmuir
MOF	Ni-MOF-74	1065	1.6	1.2	11.2	Langmuir–Freundlich
MOF	NU-100	6143	9.95	1.8	6.1	Virial Equation
COF	COF-5	1670	2.1	0.6	5.8	Dubinin–Astakhov
COF	COF-102	3472	3.3	0.9	6.2	BET Model
COF	Li-COF-366	1180	2.8	1.4	18.5	Langmuir Model

the standard BET model for multilayer adsorption [79]:

$$\frac{P}{V(P_0 - P)} = \frac{1}{V_m C} + \frac{C - 1}{V_m C} \frac{P}{P_0}$$

where V is the adsorbed volume, V_m is the monolayer capacity, C is the BET constant related to adsorption energy, and P_0 is the saturation pressure [79]. However, for functionalized COFs with enhanced binding sites, the Langmuir model becomes more appropriate due to the creation of specific high-energy adsorption sites [84].

The temperature dependence of adsorption mechanisms varies significantly among the three material classes [77]. Carbon materials show the steepest decline in capacity with increasing temperature due to weak van der Waals interactions (4–8 kJ mol^{−1}), following the exponential trend predicted by the Dubinin–Astakhov model [78,80]. MOFs with open metal sites exhibit more favorable temperature dependence due to stronger binding energies (8–15 kJ mol^{−1}), enabling substantial uptake at temperatures up to 150–200 K, with the temperature dependence following Langmuir–Freundlich isotherm behavior [81, 82]. COFs show intermediate temperature dependence, with tunable performance through functional group modifications and structural optimization. Unmodified COFs often follow BET-type isotherms, while functionalized variants shift toward Langmuir-type behavior [83,84].

The validation of theoretical adsorption models requires comprehensive comparison with experimental data across different material classes and operating conditions [77,79]. Table 3 presents a systematic comparison of hydrogen adsorption performance and binding energies for representative materials from each category, demonstrating the range of adsorption behaviors and the effectiveness of different theoretical approaches for describing experimental observations [78,81].

The experimental data reveal key trends in hydrogen adsorption behavior across material classes [78,81,83]. High surface area materials perform well at cryogenic conditions (77 K), for example NU-100 achieves an exceptional uptake of 9.95 wt% due to its ultra-high surface area of 6143 m²/g, where the virial equation approach is required to describe high-density adsorption thermodynamics [85]. However, surface area alone does not determine storage performance. Ni-MOF-74 despite, with a comparatively modest surface area, outperforms many materials at ambient temperatures, highlighting the critical role of strong binding sites. This behavior is effectively captured by the Langmuir–Freundlich model, which accounts for multi-site adsorption [81,82].

The binding energy data demonstrate the fundamental differences in adsorption mechanisms among material classes and validate the theoretical model selections [77,81]. Carbon materials consistently show binding energies in the 4–7 kJ mol^{−1} range, reflecting their reliance on weak van der Waals interactions and supporting the use of the Dubinin–Astakhov model for microporous carbons and BET models for materials with broader pore distributions [78,80]. MOFs exhibit a broader range of binding energies depending on the presence and nature of open metal sites, with Ni-MOF-74 showing significantly enhanced binding

(11.2 kJ mol^{−1}) compared to MOF-5 (4.8 kJ mol^{−1}), necessitating the use of multi-site Langmuir or Langmuir–Freundlich models to capture the heterogeneous adsorption behavior [81,82]. COFs generally show intermediate binding energies, with dramatic enhancement possible through metal functionalization as demonstrated by Li-COF-366, where the transition from BET-type to Langmuir-type behavior reflects the creation of specific high-energy adsorption sites [83,84].

Recent advances in computational modeling have enabled more sophisticated approaches to predicting and understanding hydrogen adsorption mechanisms in porous materials [85]. Grand Canonical Monte Carlo (GCMC) simulations provide molecular-level insights into adsorption behavior by explicitly modeling the interactions between hydrogen molecules and framework atoms, revealing important details about hydrogen clustering behavior, preferential adsorption sites, and the effects of framework flexibility on storage performance [85]. These simulations have become essential for validating and refining the classical adsorption models, particularly for complex MOF structures where multiple adsorption mechanisms operate simultaneously [82].

Density functional theory (DFT) calculations have become essential tools for understanding the electronic structure basis of hydrogen adsorption, particularly for identifying and characterizing strong binding sites in MOF materials [81]. DFT studies have revealed that the binding energy at open metal sites depends critically on the metal oxidation state, coordination environment, and local electrostatic field, providing guidance for rational design of enhanced storage materials and supporting the development of more accurate theoretical models [81]. Machine learning approaches are emerging as powerful tools for accelerating the discovery and optimization of hydrogen storage materials, with neural network models trained on large databases of experimental and computational data capable of predicting adsorption isotherms and identifying promising material compositions with significantly reduced computational cost compared to first-principles methods [85].

2.5.7. Nanostructured approaches and sustainable energy integration

Nanostructured materials address key limitations of conventional physical storage methods by improving hydrogen uptake under reduced pressure and temperature conditions. Compared to bulk materials, they benefit from quantum size effects, higher surface-to-volume ratios, and controlled pore architectures. For example, nanostructured MOFs with pore sizes between 0.6–1.2 nm show 40%–60% greater hydrogen uptake than their bulk counterparts. Carbon nanotubes with tailored defect densities exhibit binding energies of 8–12 kJ mol^{−1}, approaching the optimal range for ambient-temperature storage [86].

Metal–organic frameworks show strong sensitivity to pore architecture, with hydrogen storage capacity closely tied to pore size, geometry, and surface chemistry. Empirical correlations between pore diameter and uptake enable predictive design of high-performance materials.

For cage-type MOF structures, the total hydrogen adsorption capacity can be described by the empirical relationship [87]:

$$n_{\text{tot,cage}} = 0.085 \times V_p - 0.013 \times V_p^2 \quad (1)$$

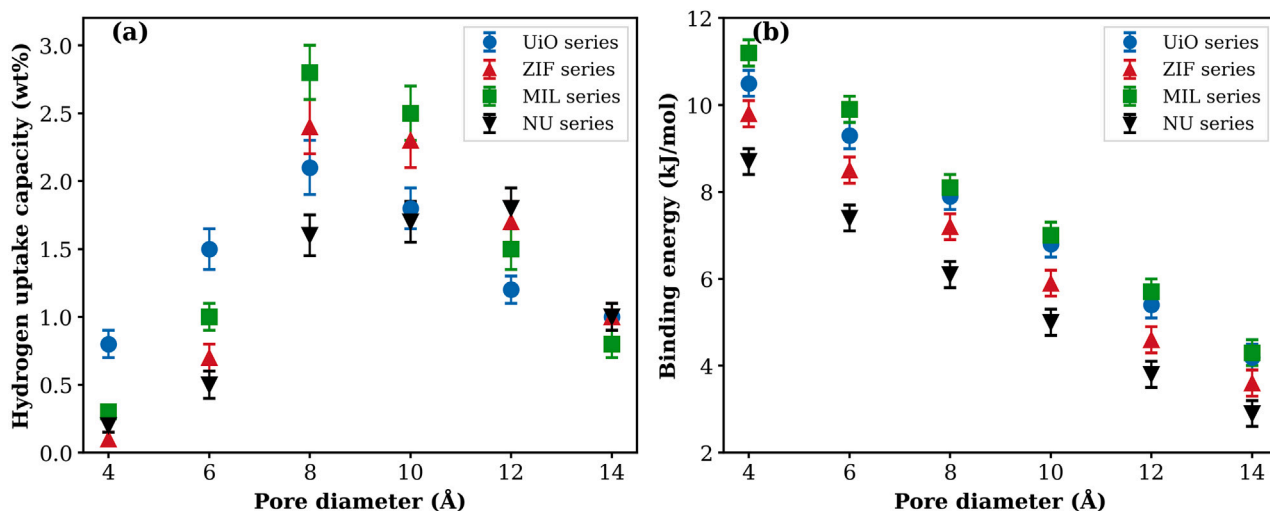


Fig. 4. Pore size effects on hydrogen storage in MOFs. (a) Hydrogen uptake capacity showing optimal performance in the 6–12 Å range across different MOF families. (b) Binding energy variation with pore diameter, demonstrating the inverse relationship between pore size and hydrogen-framework interactions. Data represent UiO series (circles), ZIF series (triangles), MIL series (squares), and NU series (inverted triangles).

while channel-type structures follow:

$$n_{\text{tot,channel}} = 0.076 \times V_p - 0.011 \times V_p^2 \quad (2)$$

where V_p is the pore volume ($\text{cm}^3 \text{g}^{-1}$) and n_{tot} represents the total hydrogen adsorption (mmol g^{-1}) at 100 bar and 77 K. These relationships reveal optimal pore volumes of $3.3 \text{ cm}^3 \text{g}^{-1}$ for cage-type and $3.5 \text{ cm}^3 \text{g}^{-1}$ for channel-type MOFs, corresponding to maximum theoretical capacities of 14.0 and 13.3 mmol g^{-1} , respectively.

An optimal pore size range of 6–12 Å balances surface area accessibility with binding energy. Pores smaller than 6 Å provide stronger interactions ($8\text{--}12 \text{ kJ mol}^{-1}$) due to overlapping potential fields, but limit accessibility and slow uptake. In contrast, pores larger than 12 Å improve accessibility but exhibit weak interactions ($3\text{--}5 \text{ kJ mol}^{-1}$), reducing retention at elevated temperatures [88]. Surface functionalization further tunes binding strength by introducing polar groups or open metal sites. The binding energy trend for divalent metals follows $\text{Mn}^{2+} > \text{Fe}^{2+} > \text{Co}^{2+} > \text{Ni}^{2+} > \text{Cu}^{2+} > \text{Zn}^{2+}$, with Mn-based frameworks reaching nearly 20 kJ mol^{-1} [82]. Decoration with metal nanoparticles provides an alternative route; CuNi@UiO-66 systems, for example, show $3.7\times$ capacity improvement (0.20 to 0.74 wt%) at room temperature via spillover effects [68].

The integration of nanostructured hydrogen storage technologies directly supports the transition to sustainable energy systems by addressing key infrastructure and efficiency challenges. These materials enable distributed storage architectures that complement renewable energy generation patterns, reducing the need for centralized, high-pressure infrastructure that requires significant capital investment and energy-intensive compression [89]. Moreover, the modular nature of nanostructured adsorbent systems facilitates rapid deployment and scaling, essential characteristics for supporting the variable output of solar and wind energy systems. The reduced energy penalties associated with moderate-pressure operation (10–50 bar vs. 700 bar) improve the overall round-trip efficiency of renewable hydrogen systems from approximately 25%–35% to 40%–50%, making hydrogen storage economically competitive with battery systems for long-duration energy storage applications [90].

2.5.8. Analyzing physical adsorption methods data

To gain a broader perspective on physisorption-based hydrogen storage, we analyzed a representative dataset from the Hydrogen Materials, Advanced Research Consortium (HyMARC) Data Hub (datahub.hymarc.org) in accordance with the user agreement. The dataset encompasses various porous material classes and operating

conditions, offering insight into the highest reported hydrogen weight percentages (wt%) achieved under different pressures and temperatures. Fig. 5 presents a frequency histogram of hydrogen weight percentages across a diverse range of adsorption-based materials. The distribution is moderately skewed, with both the mean (3.02 wt%) and median (2.80 wt%) converging around 3 wt%. While some formulations exceed 5 wt%, most fall within the 1–4 wt% range. This spread underscores the role of material structure, surface chemistry, and measurement conditions, particularly temperature and pressure, in adsorption efficiency.

Fig. 6 presents a comparative boxplot for eight material classes: Activated Carbon, Carbon Aerogel, Carbon Nanotube, Intercalated Graphite, MOF, PCN (Porous Coordination Network), Porous Organic Polymer and Templated Carbon. The data reveal distinct trends in hydrogen uptake across these materials. Carbon aerogels and activated carbons exhibit broad variability, with some samples achieving up to 5 wt%. MOFs generally display hydrogen uptake in the range of 2–4 wt%, though outliers can reach higher capacities. PCNs demonstrate some of the most promising results, with select cases exceeding 6 wt%. In contrast, templated carbons show comparatively low uptakes, often remaining below 1 wt%, but only few data points were available for this type of storage, which might not be representative.

Fig. 7 presents hydrogen uptake in a three-dimensional space, with pressure on the x-axis, temperature on the z-axis, and hydrogen weight percentage on the y-axis. The data exhibit a linear trend in hydrogen storage capacity as a function of temperature and pressure, highlighting the predictability of these materials under varying conditions. Each cluster corresponds to a distinct material class, illustrating performance variations across different operational regimes.

The trends observed in the 3D plot reveal several fundamental physical laws governing hydrogen adsorption in porous materials. The inverse relationship between temperature and adsorption capacity follows the Van't Hoff equation, where adsorption decreases exponentially with increasing temperature due to reduced hydrogen–surface interaction strength [91]. The positive correlation with pressure reflects Henry's law behavior at low pressures, transitioning to Langmuir-type saturation at higher pressures as available adsorption sites become occupied [77].

Under cryogenic conditions (77–87 K) and elevated pressures (50–100 bar), three key factors dominate hydrogen adsorption performance. Micropore accessibility becomes the primary controlling factor, as hydrogen molecules can effectively fill pores smaller than 2 nm where enhanced van der Waals interactions occur due to overlapping

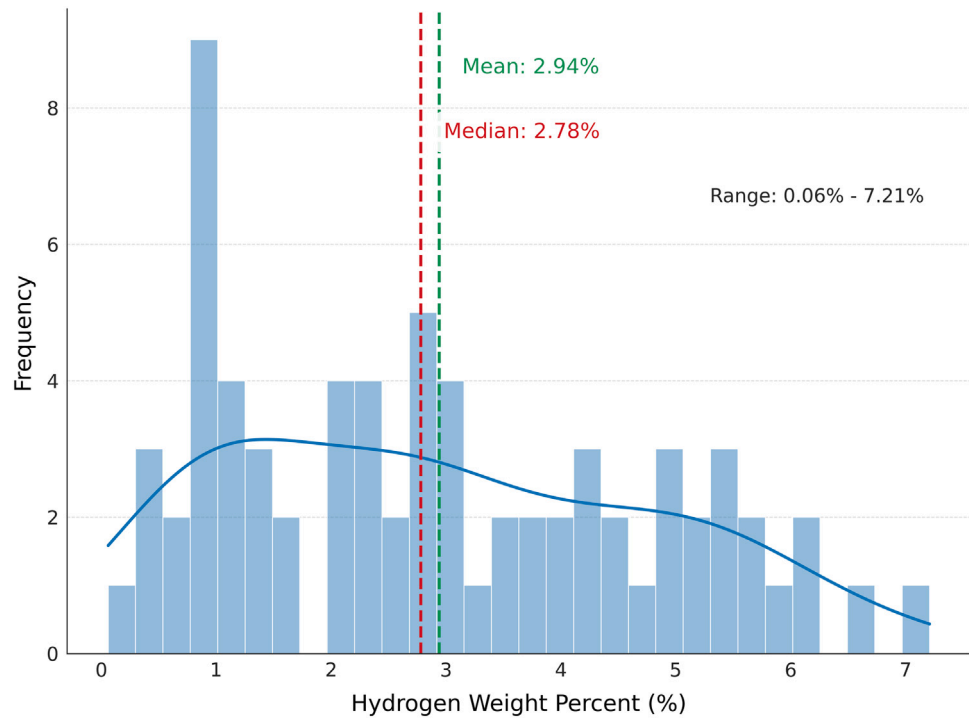


Fig. 5. Distribution of hydrogen weight percent in the dataset. The histogram (in blue) shows the frequency of observations across various hydrogen weight percentages, with a kernel density estimate (KDE) curve overlaid to illustrate the overall distribution shape. Vertical dashed lines highlight the mean (green) and median (red), to identify the central tendency and skewness of the data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

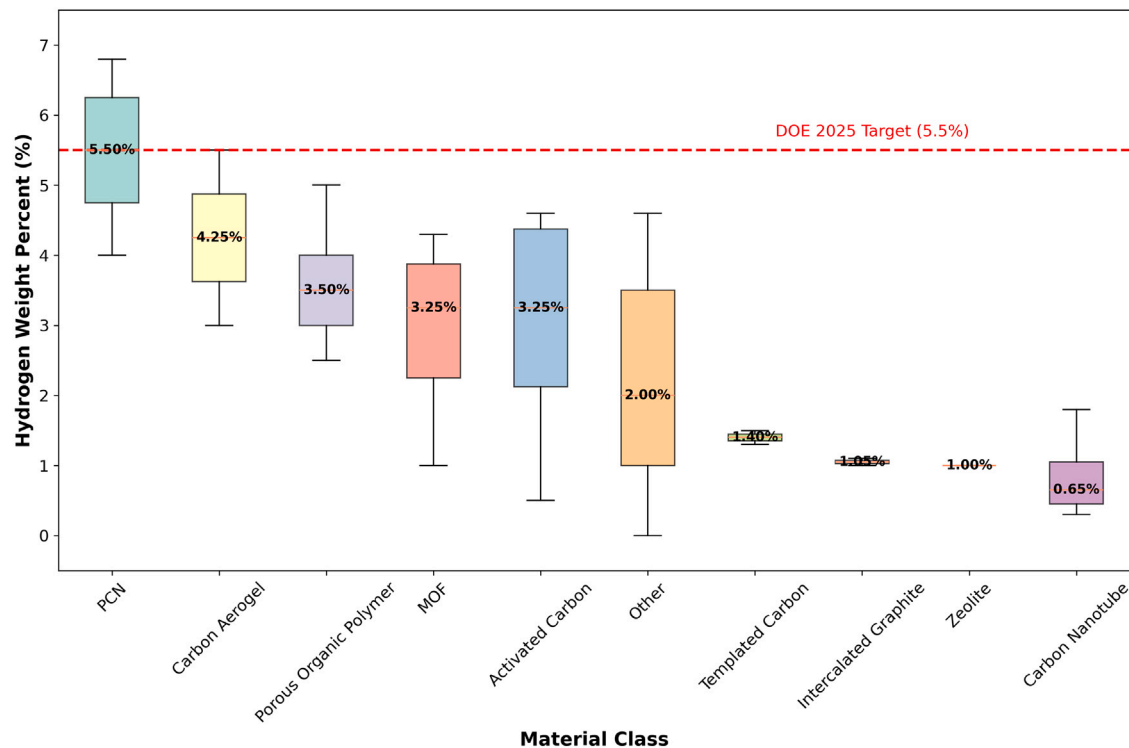


Fig. 6. Box plot comparing hydrogen weight percent across different material classes. Each box spans the interquartile range (IQR) and shows the median as a black line, while whiskers extend to represent non-outlier data. Outliers are plotted as individual points with the palette highlighting each material class.

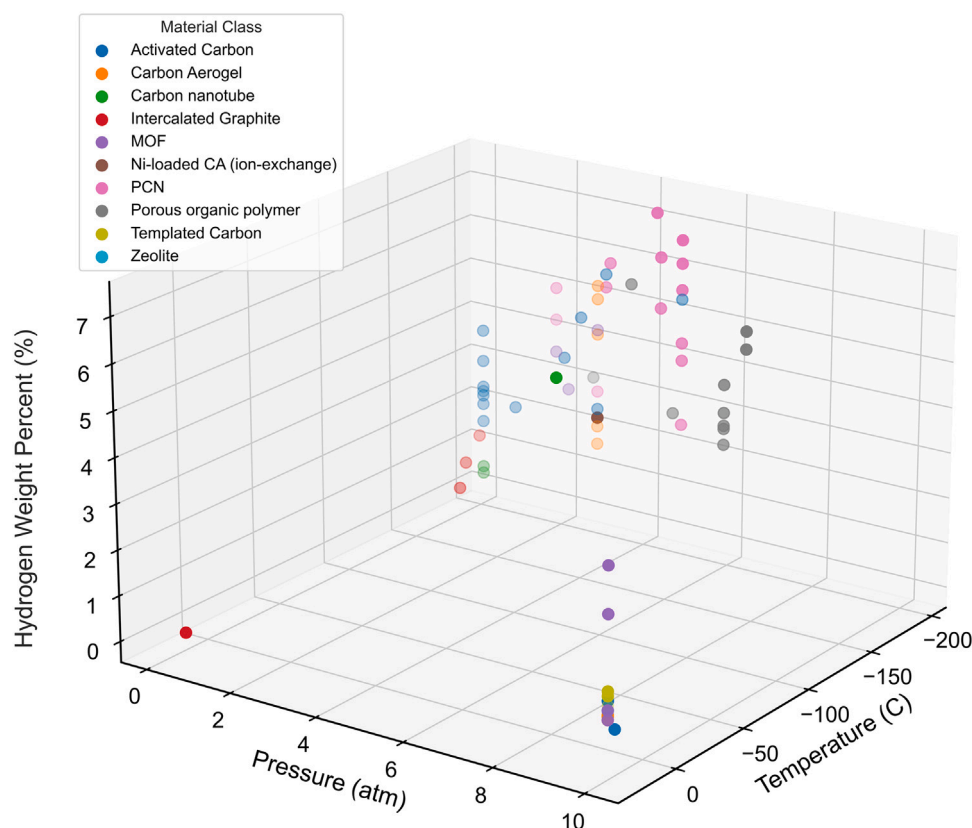


Fig. 7. 3D scatter plot of hydrogen weight percent against temperature and pressure, both on linear scales. Each material class is represented by a different color and marker shape, as specified in the legend. Axes tick labels and legends appear in black for clarity. The aspect ratio and view angle are configured to offer a balanced perspective on the 3D dataset.

potential fields from opposing pore walls [92]. Surface area utilization reaches maximum efficiency under these conditions, with materials achieving near-complete monolayer coverage before multilayer adsorption begins. Thermodynamic favorability is optimized as the reduced thermal energy allows weak physisorption interactions ($4\text{--}15\text{ kJ mol}^{-1}$) to effectively compete with molecular kinetic energy, enabling stable adsorption even on relatively weak binding sites [77].

The scatter plot data demonstrate that binding energy distribution determines material ranking under different conditions. Materials with uniform weak binding sites (activated carbons, COFs) show maximum performance only at cryogenic temperatures, while materials with strong binding sites (MOFs with open metal sites) maintain significant capacity at elevated temperatures [77]. Pore size optimization becomes critical under high-pressure conditions, where materials with pore diameters of $0.6\text{--}1.5\text{ nm}$ achieve optimal hydrogen packing density through enhanced molecule-wall interactions and efficient space utilization [92].

2.6. Summary and limitations of physical storage methods

The physical hydrogen storage methods covered here are the most mature, each with specific strengths and specific trade-offs. Compressed hydrogen enables fast refueling and simple operation, making it viable for vehicles, but it requires $10\text{--}15\%$ of hydrogen's energy for compression and has low volumetric energy density. Liquid hydrogen boosts density but needs $30\text{--}40\%$ of the energy for liquefaction and suffers from boil-off losses. Cryo-compressed storage balances density and dormancy but adds system complexity and infrastructure demands. Physisorption in porous materials is safer due to low-pressure operation but is limited by weak van der Waals interactions ($4\text{--}8\text{ kJ/mol}$). Even top-performing materials typically reach only $2\text{--}4\text{ wt\%}$ uptake at room

temperature, with $6\text{--}7\text{ wt\%}$ possible only at cryogenic temperatures ($<-150\text{ }^{\circ}\text{C}$), as shown in HyMARC data (Figs. 2–4). These technologies fall short of meeting DOE targets for gravimetric capacity ($> 5.5\text{ wt\%}$), volumetric capacity ($> 40\text{ g H}_2/\text{L}$), and room-temperature operation. Though reversible and cycle-stable, their energy costs and infrastructure needs remain major hurdles, driving the push toward chemically bonded hydrogen storage systems that may overcome these limits.

3. Chemical hydrogen storage

Chemical storage methods store hydrogen within compounds through reversible chemical reactions, releasing it when needed. This category includes metal hydrides, complex hydrides, and liquid organic hydrogen carriers (LOHCs) such as hydrogenated organic liquids and related compounds like formic acid. These methods often achieve higher volumetric storage densities and can operate at lower pressures, but they introduce challenges in reaction kinetics, energy input for hydrogen release, and reversibility.

3.1. Nanostructured materials

Nanostructured adsorbents offer significant improvements in hydrogen storage performance compared to conventional bulk materials. Their high surface area, tunable pore size, and quantum effects lead to enhanced hydrogen uptake at lower pressures and temperatures. For example, MOFs with $0.6\text{--}1.2\text{ nm}$ pores achieve $40\text{--}60\%$ greater uptake than bulk analogs, while defect-engineered carbon nanotubes reach binding energies of $8\text{--}12\text{ kJ/mol}$, close to the ideal range for ambient storage [86]. These materials also support decentralized hydrogen infrastructure by reducing dependence on high-pressure centralized systems, which are energy- and capital-intensive [89]. Their modularity allows for flexible scaling in line with variable renewable energy

output. Moderate-pressure operation (10–50 bar vs. 700 bar) improves round-trip efficiency from 25%–35% to 40%–50%, making hydrogen storage more competitive with batteries for long-duration use [90].

3.2. Metal hydrides

Metal hydrides store hydrogen by forming solid compounds (interstitial or complex phases) where hydrogen is absorbed into the metal lattice, offering high volumetric hydrogen densities and safe, relatively low-pressure storage [93]. Examples include interstitial hydrides like magnesium hydride (MgH_2), which has a high theoretical storage capacity (~ 7.6 wt%), and complex alanates like sodium alanate (NaAlH_4) or simple saline hydrides like lithium hydride (LiH). Typically, hydrogen absorption/desorption in metal hydrides occurs at moderate pressures (under 10 MPa) and, for some systems, near ambient temperatures, making them attractive for stationary storage or certain vehicular applications where weight is less critical [94].

Metal hydrides offer a promising approach to hydrogen storage, providing high volumetric densities and safe, compact storage. However, many hydrides require elevated temperatures to release hydrogen. For instance, MgH_2 desorbs hydrogen appreciably only above $\sim 300^\circ\text{C}$, posing challenges for onboard storage in vehicles [95]. Doping with transition metal catalysts can enhance kinetics and reduce desorption temperatures. A well-known example is NaAlH_4 , which, when doped with Ti-based catalysts, enables reversible hydrogen storage at $\sim 120^\circ\text{C}$, generating significant research interest [96]. Despite these advancements, metal hydrides still face limitations related to operating temperature, kinetics, stability, and practical storage capacities.

High operating temperatures remain a major drawback, with many complex hydrides requiring temperatures above 250°C and MgH_2 exceeding 300°C . Effective heat management is necessary for practical implementation. Additionally, without catalysts, absorption and desorption kinetics are slow due to hydrogen diffusion and crystal structure rearrangement. Repeated hydrogenation/dehydrogenation cycles can lead to particle coarsening or phase segregation, reducing capacity over time. For example, MgH_2 particles tend to sinter, further slowing kinetics upon cycling [97]. While volumetric capacity is high, gravimetric capacity remains a limitation, as many metal hydrides have modest hydrogen storage capacities ($< 2\text{--}3$ wt%) due to the significant mass of the metal. Only select hydrides, such as MgH_2 and AlH_3 , exceed 5 wt%, often under impractical conditions. Furthermore, some hydrides rely on expensive or heavy elements, such as LaNi_5 in NiMH batteries, which increases costs and limits large-scale adoption in transportation [98].

To address these challenges, research has focused on modifying metal hydrides through nanostructuring and catalysis. Nanostructuring enhances hydrogen absorption and desorption kinetics by increasing surface area and reducing hydrogen diffusion distances. Paskevicius et al. [99] demonstrated that ~ 7 nm MgH_2 nanoparticles exhibit a reduction in decomposition enthalpy of $2.84 \text{ kJ mol}^{-1} \text{H}_2$ compared to bulk MgH_2 ($\Delta H_{\text{nano}} = 71.22 \pm 0.49 \text{ kJ mol}^{-1} \text{H}_2$ vs. $\Delta H_{\text{bulk}} = 74.06 \pm 0.42 \text{ kJ mol}^{-1} \text{H}_2$), along with a reduction in reaction entropy ($\Delta S_{\text{nano}} = 129.6 \pm 0.8 \text{ J mol}^{-1} \text{H}_2 \text{K}^{-1}$ vs. $\Delta S_{\text{bulk}} = 133.4 \pm 0.7 \text{ J mol}^{-1} \text{H}_2 \text{K}^{-1}$). This thermodynamic alteration results in a lower hydrogen equilibrium desorption temperature at 1 bar, with $T_{1 \text{ bar}}$ reduced by $\sim 6^\circ\text{C}$ to $276.2 \pm 2.4^\circ\text{C}$ compared to bulk MgH_2 at $281.8 \pm 2.2^\circ\text{C}$.

Additionally, it can alter thermodynamics, potentially lowering the temperature required for hydrogen release [100]. For instance, MgH_2 nanopowders begin desorbing hydrogen at lower temperatures (250°C or less) compared to bulk MgH_2 ($\sim 350^\circ\text{C}$) [101]. Varin et al. [102] reported that bulk MgH_2 requires temperatures of $350\text{--}400^\circ\text{C}$ for significant hydrogen desorption, while Aguey-Zinsou et al. [103] demonstrated that 5 nm MgH_2 nanoparticles synthesized via chemical reduction begin desorbing hydrogen at temperatures as low as $150\text{--}200^\circ\text{C}$, representing a reduction of approximately $150\text{--}200^\circ\text{C}$ in the onset desorption temperature. Various synthesis methods have been explored

to produce hydride nanostructures. Mechanical ball milling is a top-down approach that reduces micrometer-sized hydride powders to nanometer scales, significantly improving kinetics. For example, ball-milled MgH_2 with Nb_2O_5 catalysts has demonstrated faster hydrogen absorption and desorption [97]. Mechanochemical synthesis enables the direct formation of hydride nanoparticles during milling. Paskevicius et al. (2010) synthesized ~ 7 nm MgH_2 nanoparticles via a reaction between LiH and MgCl_2 , achieving reduced decomposition enthalpy and enhanced kinetics [99]. Chemical reduction methods, such as using Grignard reagents, allow precise control over nanoparticle size and surface properties. Aguey-Zinsou & Ares-Fernández (2007) synthesized ~ 5 nm MgH_2 colloids through chemical reduction, significantly lowering dehydrogenation temperatures compared to bulk MgH_2 [103]. Beyond nanoparticles, nanowires and nanorods have been investigated for hydrogen storage. Magnesium nanowires ($\sim 30\text{--}50$ nm diameter) produced via chemical vapor deposition exhibit improved hydrogen absorption and desorption kinetics due to internal stresses and shorter diffusion pathways [104]. Similarly, Pang et al. (2014) fabricated Mg_2FeH_6 nanorods using a mechanical-force-driven vapor deposition approach, demonstrating enhanced hydrogen release properties [105].

Another approach involves incorporating hydrides into nanostructured scaffolds to prevent particle agglomeration, enhance heat transfer, and maintain nanoscale features during cycling. Carbon-based materials, such as graphene and porous carbon, are particularly effective due to their high surface area and chemical stability. For instance, MgH_2 nanoparticles (~ 5 nm) dispersed on graphene achieve high loading (~ 75 wt% MgH_2 in composite) while maintaining fast hydrogenation at 50°C and significant hydrogen release at 200°C , far lower than bulk MgH_2 [106]. Similarly, NaAlH_4 confined within a carbon aerogel framework exhibits improved reversibility and kinetics [107]. Metal-organic frameworks (MOFs) and other porous hosts have also been explored as nanoconfinement strategies. Infiltrating LiBH_4 into MOF or oxide matrices can lower melting points and improve cycling performance, although full reversibility remains a challenge [108]. Functionalized supports can provide additional catalytic effects, enhancing hydrogen uptake and release through spillover mechanisms [109]. Through these nanostructuring and composite approaches, substantial improvements have been achieved, with many hydrides now exhibiting lower hydrogen release temperatures and significantly enhanced kinetics. However, challenges remain in scaling up nanoscale hydrides while ensuring long-term stability and safety. The potential for pyrophoric behavior in nanopowders and the high cost of nanoscale processing must be addressed. Despite these hurdles, ongoing advancements in nanostructured hydrides represent important steps toward making metal hydrides a viable hydrogen storage solution for practical applications [110].

Another critical aspect is the stability of metal hydrides under actual storage conditions, governed by three critical factors: temperature cycling effects, pressure variations, and mechanical degradation from repeated hydrogen absorption–desorption cycles [111]. Temperature cycling between ambient and operating conditions ($20\text{--}400^\circ\text{C}$) induces thermal stress that can cause particle decrepitation and surface oxidation, particularly affecting MgH_2 , which operates at elevated temperatures [112]. Pressure cycling between vacuum and storage pressures ($1\text{--}100$ bar) creates mechanical stress that leads to particle pulverization and loss of inter-particle contact, reducing effective thermal conductivity and hydrogen transport kinetics [113]. Cycle life performance varies significantly among different metal hydride systems, with degradation mechanisms including capacity fade, kinetic deterioration, and structural changes. LaNi_5 -based alloys demonstrate excellent cycling stability with minimal capacity loss ($< 5\%$) over $1500\text{--}2000$ cycles at moderate temperatures ($20\text{--}60^\circ\text{C}$), while MgH_2 -based systems show more pronounced degradation ($10\text{--}20\%$ capacity loss) over $100\text{--}500$ cycles due to higher operating temperatures and associated thermal stress [111,114]. Advanced composite materials incorporating graphite or carbon additives show improved stability,

Table 4

Stability test data for representative metal hydride materials under actual storage conditions. Capacity retention values are relative to initial performance after the specified number of cycles.

Material	Operating temp (°C)	Pressure (bar)	Cycle life	Capacity retention	Key degradation factors	Reference
LaNi ₅	20–60	1–20	1500–2000	95% (after 1500 cycles)	Surface oxidation, decrepitation	[114]
LaNi _{4.7} Al _{0.3}	20–60	1–20	2000	85% (after 2000 cycles)	Reduced compared to LaNi ₅	[114]
MgH ₂	300–400	1–30	100–500	80%–90% (after 100 cycles)	Thermal stress, sintering	[112]
MgH ₂ –Ni	250–350	1–30	450+	85% (after 450 cycles)	Improved vs. pure MgH ₂	[115]
TiFe	20–100	5–50	1000+	90% (after 1000 cycles)	Moderate degradation	[116]
Hydralloy C5–ENG	20–80	1–30	1000	95% (after 1000 cycles)	Thermal conductivity loss	[111]

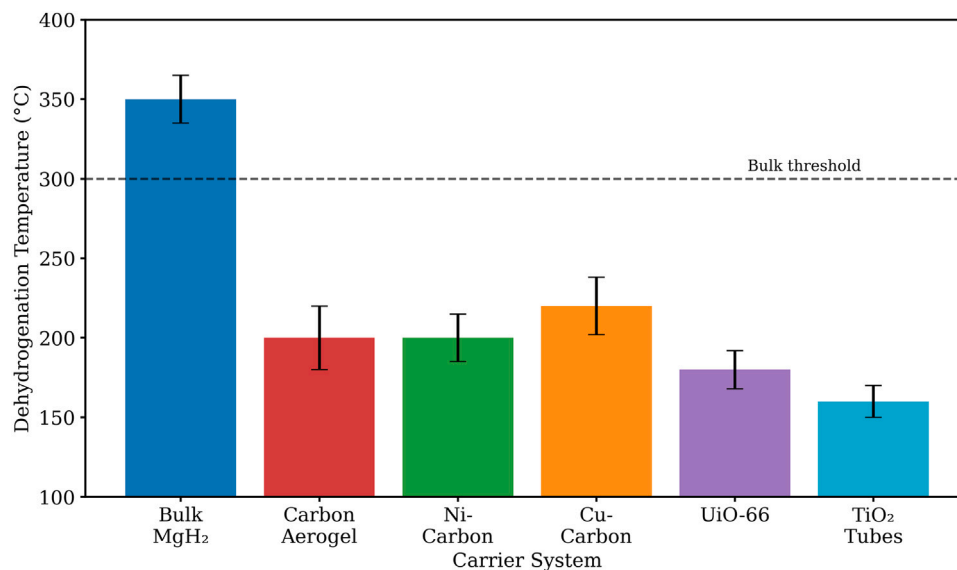


Fig. 8. Dehydrogenation temperature comparison for MgH₂ in different carrier systems. All nanoconfined systems show significant temperature reductions compared to bulk MgH₂ (> 300 °C), with reductions ranging from 100–180 °C depending on the carrier material and pore structure.

with thermal conductivity stabilizing after initial degradation during the first 250 cycles [111]. To provide a comprehensive comparison of stability performance across different material classes, Table 4 summarizes experimental data from long-term cycling studies conducted under realistic storage conditions. These data demonstrate the wide variation in cycle life and capacity retention among different metal hydride systems.

3.2.1. Nanoconfinement in metal hydride systems

Nanoconfining metal hydrides within porous scaffolds simultaneously lowers thermodynamic stability and improves kinetics, addressing key limitations of bulk hydrides. These effects depend on scaffold type, pore size, and hydride–scaffold interactions. Magnesium hydride (MgH₂) is the benchmark system for nanoconfinement studies due to its high theoretical capacity (7.6 wt%) and well-known bulk behavior. In bulk, MgH₂ requires dehydrogenation temperatures above 300 °C and shows slow kinetics. Confinement within porous hosts significantly reduces these barriers, enhancing both kinetics and thermodynamics. (See Fig. 8.)

Carbon-based scaffolds offer chemically inert, high-surface-area environments with tunable porosity. Confinement of MgH₂ in carbon aerogels reduces dehydrogenation temperatures to ~200 °C—a 150 °C drop from bulk values [117]. MgH₂ interacts with carbon primarily via van der Waals forces, ensuring good reversibility and moderate destabilization. Metal-decorated carbons add catalytic functionality; Ni-doped systems achieve complete hydrogen release within 30 min at 250 °C. Metal–organic frameworks (MOFs) provide ordered pore networks with tunable chemistry. MgH₂ confined in UiO-66 shows dehydrogenation at ~180 °C, with the rigid framework maintaining nanoparticle

dispersion and preventing sintering [118]. Zirconium nodes introduce weak Lewis acid sites that facilitate H₂ dissociation and recombination. Oxide-based supports, particularly TiO₂ nanotubes, yield the greatest temperature reductions, with dehydrogenation near 160 °C. Strong metal–support interactions enhance destabilization but often reduce reversibility. Interfacial phase formation can degrade capacity over extended cycling [119].

3.2.2. Kinetic enhancement mechanisms

Nanostructuring provides dramatic kinetic improvements through multiple pathways including reduced diffusion distances, increased surface-to-volume ratios, and enhanced nucleation site density. The quantitative relationship between particle size and kinetic performance follows power law scaling relationships that enable predictive design of fast-kinetics systems. (See Fig. 9.)

Hydrogen sorption kinetics in metal hydrides are strongly size-dependent. The absorption rate constant follows $k_{\text{abs}} \propto d^{-1.5}$, and desorption scales as $k_{\text{des}} \propto d^{-1.2}$, where d is the particle diameter [120]. These relationships predict substantial rate enhancements, up to 100× for absorption and 30× for desorption, when particle sizes are reduced from 100 nm to 5 nm. This improvement is attributed to reduced diffusion lengths and higher surface-area-to-volume ratios. In bulk MgH₂, hydrogen diffusion through thick hydride layers limits reaction rates, resulting in apparent activation energies of 160–180 kJ/mol^{−1}. Nanoconfinement lowers these values to 110–140 kJ/mol^{−1} and increases pre-exponential factors by 2–3 orders of magnitude due to a higher density of accessible active sites [121]. Temperature-dependent

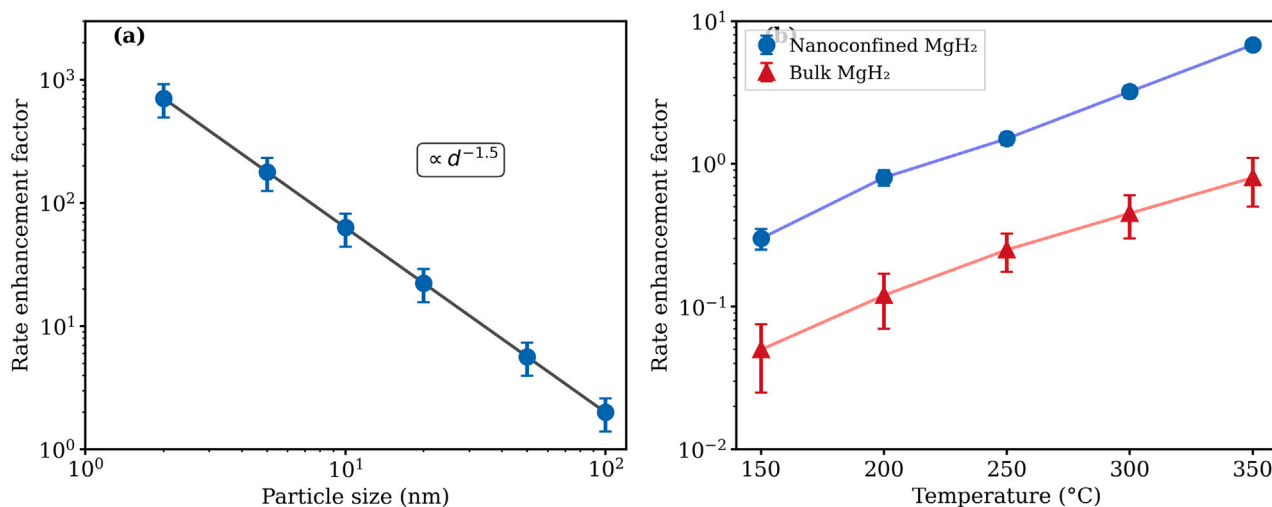


Fig. 9. Kinetic enhancement in nanostructured systems. (a) Rate enhancement factor versus particle size showing power law scaling ($\propto d^{-1.5}$) for absorption processes. (b) Temperature-dependent performance comparing nanoconfined and bulk MgH₂ systems, demonstrating superior kinetics across the operational temperature range.

studies further show that nanoconfined MgH₂ maintains superior kinetics across the operational range, with rate enhancement factors increasing from 0.8–1.5 at 200 °C to 3–7 at 300 °C [122].

Combining nanostructuring with catalytic doping yields synergistic effects beyond those of either strategy alone. Transition metal additives such as Ni, Ti, and V reduce activation barriers by enabling alternate reaction pathways, while nanostructuring improves catalyst dispersion and accessibility. This combination enables full hydrogen desorption within 5–10 min at temperatures 100–150 °C lower than bulk MgH₂ [123]. These advances position nanoconfined hydride systems as promising candidates for mobile hydrogen storage. By tailoring scaffold materials and controlling particle size, both thermodynamic and kinetic properties can be optimized. Ongoing work aims to refine this balance to achieve high capacity, fast kinetics, and long-term cycling stability [124].

3.2.3. Nanostructure–catalyst synergy in metal hydrides

The integration of nanostructuring and catalytic doping has significantly advanced metal hydride performance, with synergistic effects that surpass the contributions of either approach alone. This strategy has been particularly effective in enabling reversible hydrogen storage at practical temperatures with improved cycling stability. A key example is the titanium-doped sodium alanate (NaAlH₄) system, where Ti-based catalysis combined with nanoscale dispersion has produced near-commercial performance metrics [125].

In Ti-doped NaAlH₄, the catalytic mechanism involves a progressive transformation of Ti species during cycling: Ti³⁺ → amorphous Ti⁰ → AlTi phases → Al₃Ti intermetallics. These transformations result in uniformly dispersed active sites, ~30 nm in size, embedded within the aluminum matrix. These sites catalyze hydrogen dissociation and facilitate spillover processes. Rehydrogenation proceeds via (1) H₂ dissociation on Ti sites, (2) spillover and alane (Al_xH_y) formation, and (3) reconstruction of NaAlH₄. This mechanistic insight provides a framework for designing more active and stable catalyst systems [125]. Combining nanostructuring with Ti-based catalysis yields performance gains well beyond those achieved by either method alone. Nanostructured MgH₂ can lower dehydrogenation temperatures by 100–150 °C, while Ti-catalyzed bulk NaAlH₄ enables reversible storage near 120 °C. When both strategies are applied together, optimized systems reach dehydrogenation temperatures as low as 80 °C with full reversibility [126], a reduction of over 220 °C compared to uncatalyzed bulk NaAlH₄. This synergy results from improved catalyst dispersion and

accessibility via nanostructuring, along with shorter diffusion paths and increased surface area that enhance catalytic efficiency.

Recent advances demonstrate the potential of combining nanostructuring with tailored catalyst design for high-performance hydrogen storage. Ren et al. developed TiH₂ nanoplates supported on graphene (NP-TiH₂@G), achieving 5 wt% reversible capacity with full dehydrogenation at 80 °C and rehydrogenation at 30 °C under 100 atm H₂ [126]. The nanoplates (~50 nm lateral, 15 nm thick) are synthesized sonochemically, with graphene serving as a template, agglomeration barrier, and thermal conductor. With 7 wt% TiH₂ loading (~70% catalyst by mass), this system outperforms previous designs. In parallel, Yuan et al. reported N-doped carbon-coated Ti₃C₂ MXene catalysts with onset dehydrogenation at 85 °C and less than 5.5 wt% capacity loss over 10 cycles [127]. The MXene offers high conductivity and functional group density, while nitrogen doping introduces additional active sites and tunes hydrogen binding. Further enhancement was achieved by integrating CeF₃ nanoparticles, reducing the onset temperature to 87 °C and improving cycling stability through complementary catalytic mechanisms.

The data in Table 5 highlights major advances in the cycling stability of Ti-catalyzed NaAlH₄ systems. The nano-Ti@C catalyst shows exceptional durability, with only 0.07 wt% capacity loss over 100 cycles (degradation <0.001%/cycle) [129]. This is attributed to carbon coating, which prevents sintering and maintains catalyst dispersion. Similarly, the Al₆₅Cu₂₀Fe₁₅-LBMACF system demonstrates excellent stability (~0.06 wt% loss over 25 cycles), indicating the potential of quasicrystalline catalysts [128]. Degradation in Ti-based systems is primarily driven by sintering, formation of inactive intermetallics, surface oxidation, and mechanical fracturing. These challenges are mitigated through protective coatings, optimized synthesis, and support materials that stabilize active sites.

Quantitative studies confirm that combining nanostructuring with catalysis yields synergistic improvements. While nanostructuring or catalysis alone enhances kinetics by 10–50× and 5–20×, respectively, their combination often achieves 100–500× improvement [121]. Nanostructuring shortens diffusion paths and increases surface area, while catalysis lowers activation energy barriers. Recent directions include single-atom catalysts, core-shell architectures, and multifunctional materials that integrate kinetic and thermodynamic tuning. Machine learning is also being employed to accelerate optimization of catalyst composition and nanostructure design, moving toward systems that meet DOE targets for reversible, high-capacity hydrogen storage.

Table 5Latest cycle stability data for Ti-Catalyzed NaAlH₄ systems.

Catalyst system	Dehydrogenation temperature (°C)	Cycles tested	Capacity retention (%)	Degradation rate (%/cycle)	Key performance features	Reference
NP-TiH ₂ @G (7 wt%)	80	Multiple	5 wt% reversible	Not specified	• Full dehydrogenation at 80 °C • Rehydrogenation at 30 °C • Superior to all literature results	[126]
Ti ₃ C ₂ /NC MXene	85	10	>94.5	<0.55	• N-doped carbon coating • 3.0 wt% hydrogen uptake • Enhanced kinetics	[127]
CeF ₃ + Ti ₃ C ₂ MXene	87	10	>94.5	<0.55	• Synergistic catalytic effect • Record stability and reversibility • Multi-component optimization	[127]
Al ₆₅ Cu ₂₀ Fe ₁₅ -LBMACF	150	25	>98.8	~0.05	• Minimum degradation • Long-term stability • Quasicrystalline structure	[128]
Nano-Ti@C	Not specified	100	>99.93	<0.001	• Highly stable cyclability • Only 0.07 wt% loss after 100 cycles • Metallic Ti catalyst	[129]
Ti ₂ C MXene (in-situ)	Reduced onset	Not specified	Enhanced	Not specified	• In-situ doping via ball milling • 2D MXene structure • Moderate synthesis conditions	[130]

3.3. Complex hydrides

Complex hydrides are chemical compounds where hydrogen is bonded in complex anions (such as [BH₄][−], [AlH₄][−]) often paired with light cations. They offer some of the highest known gravimetric hydrogen capacities. For example, lithium borohydride (LiBH₄) contains 18.5 wt% hydrogen (theoretical) and ammonia borane (NH₃BH₃) ~ 19.6 wt%. The allure of complex hydrides is tempered by their thermodynamic stability and kinetic sluggishness, which typically make hydrogen release difficult [131].

A representative complex hydride is ammonia borane (NH₃BH₃). It releases hydrogen in a three-step decomposition starting around 110 °C and finishing by ~ 150 °C (for ~ 12 wt% H₂ released), but the spent material is a polyborazine polymer that is difficult to rehydrogenate [19]. Baitalow et al. [132] characterized the dehydrogenation of NH₃BH₃ in three distinct steps: first at 70–112 °C (release of the first equivalent of H₂), second at 110–150 °C (release of the second equivalent), and third at >150 °C (release of the third equivalent). While the theoretical hydrogen content is 19.6 wt%, approximately 12–13 wt% is practically accessible below 150 °C through the first two steps. Kondrat'ev et al. [133] determined that the activation energy for hydrogen evolution from solid NH₃BH₃ is 94 ± 11 kJ mol^{−1} in the temperature range of 327–351 K, with a measured sublimation enthalpy of 76.3 ± 3.0 kJ mol^{−1} at 357 K. Significant research has gone into making ammonia borane more practical, such as using nano-confined AB in porous hosts or reactive additives to promote complete hydrogen release at lower temperatures, similar to the study by Chen et al. (2024). Despite progress, AB is seen more as a one-time use chemical hydrogen source (for fuel cells in portable power, for example) rather than a rechargeable storage medium [134].

Metal Borohydrides (MBH₄): LiBH₄, NaBH₄, Mg(BH₄)₂, etc., have very high hydrogen content but very high stability. LiBH₄ releases hydrogen in multiple steps from 320 °C upward, and direct rehydrogenation of LiH + B to LiBH₄ requires > 600 °C and > 35 MPa H₂, not feasible in onboard systems. Doping and reactive hydride composites have had some success (e.g., LiBH₄ + MgH₂ mixtures show lower combined dehydrogenation temperature due to mutual destabilization). Mg(BH₄)₂ has attracted attention because it can reversibly release ~ 9–14 wt% H₂ under 200–300 °C with appropriate catalysts, but full reversibility is still not achieved [135].

Alanes (Aluminum hydrides): NaAlH₄ (sodium alanate) is the classic case where Ti-catalysts (e.g., TiCl₃) enabled a reversible capacity of ~ 4 wt% at ~ 120 °C (vs. 3.7 wt% theoretical). This breakthrough by

Bogdanovic in 1997 showed that complex hydrides could be tuned for reversibility, spurring efforts into other alanes and borohydrides [96]. Still, the capacity of NaAlH₄ is modest and the need for careful catalyst addition and O₂-free handling are issues.

Complex hydrides often operate at moderate to high pressures to uptake hydrogen (some requiring > 5 MPa). Many are not truly reversible under practical conditions, making them more akin to chemical hydrogen carriers (to be discussed below) where the spent fuel is regenerated off-board. Research continues into novel complex hydrides (e.g., mixtures like LiBH₄–MgH₂, a destabilized system that shows improved kinetics [136]; or transition-metal borohydrides with potentially lower stability). The key challenges remain high desorption temperatures, slow kinetics of hydrogen release, and in some cases the irreversibility or by-product formation during decomposition [137].

Advanced borohydride systems have experienced renewed interest due to breakthrough developments in catalyst design and system optimization. Magnesium borohydride systems now achieve gravimetric capacities of 14.9 wt% through additive enhancement, while sodium borohydride offers controlled hydrogen release through improved hydrolysis catalysts [138,139]. These systems provide high storage densities suitable for mobile applications while addressing previous challenges related to byproduct management and system regeneration. Recent advances have focused on developing more efficient catalysts and simplified system designs that reduce operational complexity.

3.4. Analysis of metal & complex hybrids storage data

To gain a comprehensive understanding of metal hydride performance, we analyzed a broad dataset of known metal hydrides obtained from the Hydrogen Materials, Advanced Research Consortium (Hy-MARC) Data Hub (datahub.hymarc.org) in accordance with the user agreement. The dataset includes hydrogen weight percentages (wt.%) for a wide range of metal hydrides, encompassing different classes such as AB, A₂B, AB₂, AB₅, metal-intercalated carbons (MIC), solid solutions (SS), magnesium hydrides (Mg), and complex hydrides. A statistical analysis of over 1300 samples from the literature reveals hydrogen storage capacities ranging from < 0.1 wt% to ~ 15 wt% under various conditions, as summarized in Table 6. However, the majority of investigated hydrides exhibit capacities below 3 wt%, which remains insufficient to meet DOE targets for onboard hydrogen storage systems. The dataset also reflects significant variation in temperature and pressure conditions, highlighting the diversity of material classes and testing protocols used in these studies.

Table 6
Statistical description of metal hydrides hydrogen storage data.

	Hydrogen Weight Percent (%)	Temperature (°C)	Pressure (MPa)
count	1300.000000	1300.000000	1300.000000
mean	1.948169	99.814615	2.144502
std	1.550483	112.143310	2.426366
min	0.100000	85.000000	0.000100
25%	1.100000	25.000000	0.400000
50%	1.500000	40.000000	1.000000
75%	2.100000	150.000000	3.000000
max	14.900000	405.000000	10.000000

This study includes various categories of metal hydrides, each with distinct hydrogen storage properties and operational constraints. AB, A₂B, AB₂, AB₃ hydrides, which are intermetallic compounds, are well known for their reversible hydrogen absorption and desorption capabilities. They generally operate under moderate temperatures and pressures but offer relatively low hydrogen storage capacities, typically below 2 wt%. Metal-intercalated carbons (MICs) incorporate metals into carbon structures to enhance hydrogen adsorption properties. However, their storage capacities remain modest and require specific conditions for optimization. Solid solution (SS) hydrides consist of metals dissolved into one another, forming a single-phase alloy. While their hydrogen storage capacities are relatively low, they benefit from improved kinetics and stability. Magnesium hydrides (Mg) stand out due to their higher hydrogen storage capacities, often exceeding 5 wt%. However, their practical application is limited by the need for elevated temperatures, typically above 300°C, for hydrogen release. Complex hydrides, including alanates and borohydrides, exhibit some of the highest hydrogen storage capacities, with some materials exceeding 10 wt%. Despite their potential, these hydrides often require high temperatures and pressures for hydrogen uptake and release, and they suffer from slow kinetics and cycling stability issues.

Fig. 10 illustrates the distribution of hydrogen weight percent (H₂ wt%) among metal hydrides from this dataset. The histogram reveals that most materials cluster in the 0.5–3 wt% range, with a long tail extending to higher capacities for a few outliers (such as certain complex borohydrides). This indicates that while some materials (e.g., LiBH₄, AlH₃) exhibit high theoretical capacities, practical realizations often fall short due to issues like incomplete reversibility or requiring extreme conditions. Notably, current material capacities remain below the DOE's ultimate gravimetric target ($\approx$ 6.5 wt%) in reversible form, highlighting the need for further innovation.

Fig. 11 compares hydrogen storage capacities across the different material categories. The boxplot provides a snapshot of both median performance and variability within each class where hydrogen storage capacity (wt%) is plotted against operating temperature and pressure for various hydride classes, including AB₃ alloys, AB₂ alloys, and complex hydrides. This visualization highlights the trade-offs between capacity and operational conditions. For instance, intermetallic AB₃ and AB₂ hydrides, such as LaNi₅-based and TiFe-based compounds, function near ambient temperature and pressure but offer low capacities, typically below 2 wt%. In contrast, complex hydrides like alanates and borohydrides can achieve storage capacities exceeding 5 wt%, though only at high temperatures (150–350 °C) and with sluggish kinetics.

Intermetallic hydrides such as LaNi₅ and FeTi are known for their reversible hydrogen absorption and desorption. They typically operate under mild temperatures, ranging from –20 to 60°C, and near-ambient pressures of 1–10 bar. Their stability and rapid kinetics make them suitable for applications such as NiMH batteries and stationary hydrogen storage. However, their low hydrogen capacities prevent their use in mobile applications where weight efficiency is critical. Another category, solid-solution hydrides, includes metals that form continuous solid solutions with hydrogen, such as palladium hydride. While these materials exhibit strong hydrogen absorption characteristics, their low

gravimetric capacity confines their use to niche applications, such as palladium membranes. Magnesium hydrides, particularly MgH₂, offer significantly higher hydrogen storage capacities, with a theoretical maximum of approximately 7 wt%. However, they require temperatures between 300 and 400 °C for hydrogen release, which limits their practical viability. Additionally, they exhibit slow absorption and desorption kinetics unless modified through nanostructuring or catalytic doping. Despite these drawbacks, magnesium hydrides remain attractive due to their abundance, low cost, and high volumetric hydrogen density of around 110 kg H₂/m³.

Complex hydrides, including sodium alanate (NaAlH₄), lithium borohydride (LiBH₄), and ammonia borane (NH₃BH₃), possess some of the highest known hydrogen storage capacities, with certain borohydrides exceeding 10 ~ wt%. However, these materials often require elevated temperatures for hydrogen release, typically above (120 °C) for alanates with catalysts and over 250 °C for borohydrides. Many borohydrides also suffer from irreversible decomposition or sluggish kinetics. For example, LiBH₄ releases hydrogen through multiple steps up to $\sim$ 400 °C, and its rehydrogenation requires even harsher conditions, making it impractical for reversible storage without catalysts or nanoconfinement strategies.

A promising avenue for improving hydrogen storage performance involves nanoconfinement, where hydrides are embedded within nanoporous scaffolds such as carbon aerogels or metal-organic frameworks (MOFs). This strategy enhances kinetics and can reduce reaction enthalpies, facilitating hydrogen release at lower temperatures. For instance, NaAlH₄ melt-infiltrated into a carbon aerogel exhibited dehydrogenation at approximately 150 °C, significantly lower than bulk NaAlH₄, while maintaining good reversibility [140]. These results highlight the potential of nanostructuring to overcome some of the intrinsic limitations of metal hydrides.

While complex and magnesium hydrides offer higher hydrogen storage capacities, their practical application remains constrained by their demanding operational conditions, including high temperatures, slow kinetics, and, in some cases, irreversibility. In contrast, intermetallic hydrides, though limited in capacity, operate under milder conditions and demonstrate excellent cycle stability. The balance between storage capacity and operational feasibility is a critical consideration in advancing hydrogen storage technologies. The results of Fig. 12 supports the above claims about the abilities of metal hybrids for storing hydrogen under different conditions.

3.5. Liquid organic hydrogen carriers (LOHCs)

Liquid Organic Hydrogen Carriers (LOHCs) enable reversible hydrogen storage through hydrogenation and dehydrogenation reactions, allowing for safe and efficient hydrogen transport in liquid form under ambient conditions [141]. These compounds absorb hydrogen through exothermic hydrogenation, forming saturated molecules, and later release it via endothermic dehydrogenation, regenerating the original unsaturated compound. Functioning as a “liquid battery” for hydrogen, LOHCs store hydrogen chemically and release it when needed. Ideal LOHCs should exhibit high hydrogen storage capacities, favorable thermodynamics and kinetics, chemical and thermal stability, low toxicity, and economic viability [142].

Several LOHC systems have been extensively studied. N-ethylcarbazole (NEC) was among the earliest demonstrated LOHCs, with a theoretical hydrogen capacity of $\sim$ 5.8 wt%. Hydrogenation of NEC forms perhydro-N-ethylcarbazole, which requires dehydrogenation at $\sim$ 180–210 °C over a catalyst such as Pd on alumina to release hydrogen, with NEC being fully regenerated [141,143]. Peters et al. [144] determined that optimal dehydrogenation occurs at 180–200 °C with suitable catalysts, with a reaction enthalpy of approximately 50–53 kJ mol^{–1} H₂. This relatively high dehydrogenation enthalpy necessitates significant heat input for hydrogen release, which remains a challenge for practical applications. This relatively high

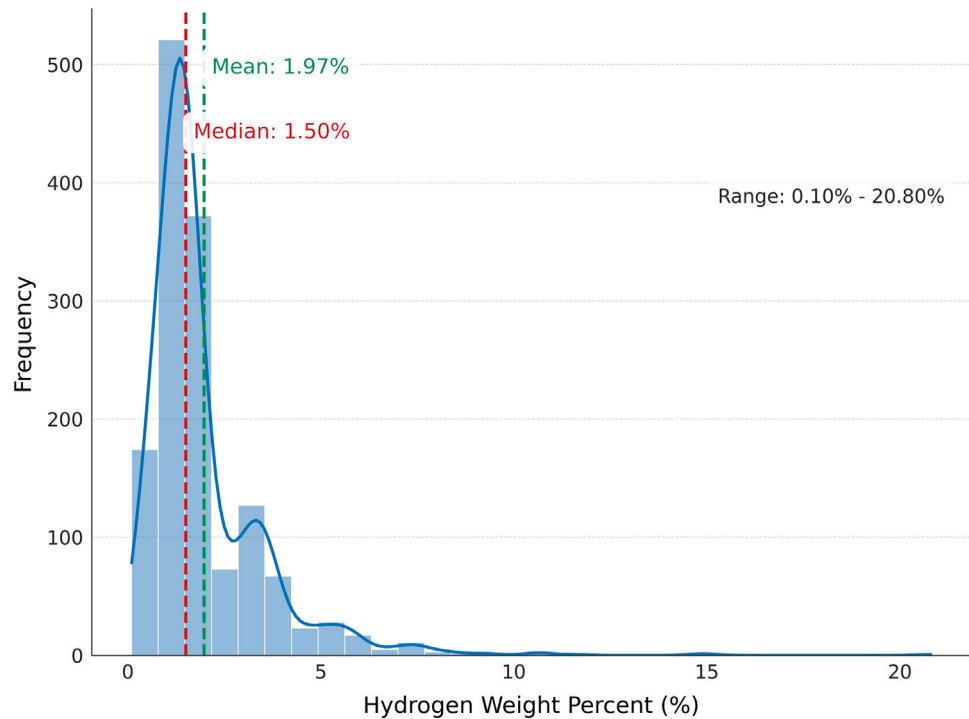


Fig. 10. Distribution of hydrogen weight percent in the dataset. The histogram (in blue) shows the frequency of observations across various hydrogen weight percentages, with a kernel density estimate (KDE) curve overlaid to illustrate the overall distribution shape. Vertical dashed lines highlight the mean (green) and median (red). (Data compiled from DOE Metal Hydride databases and literature.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

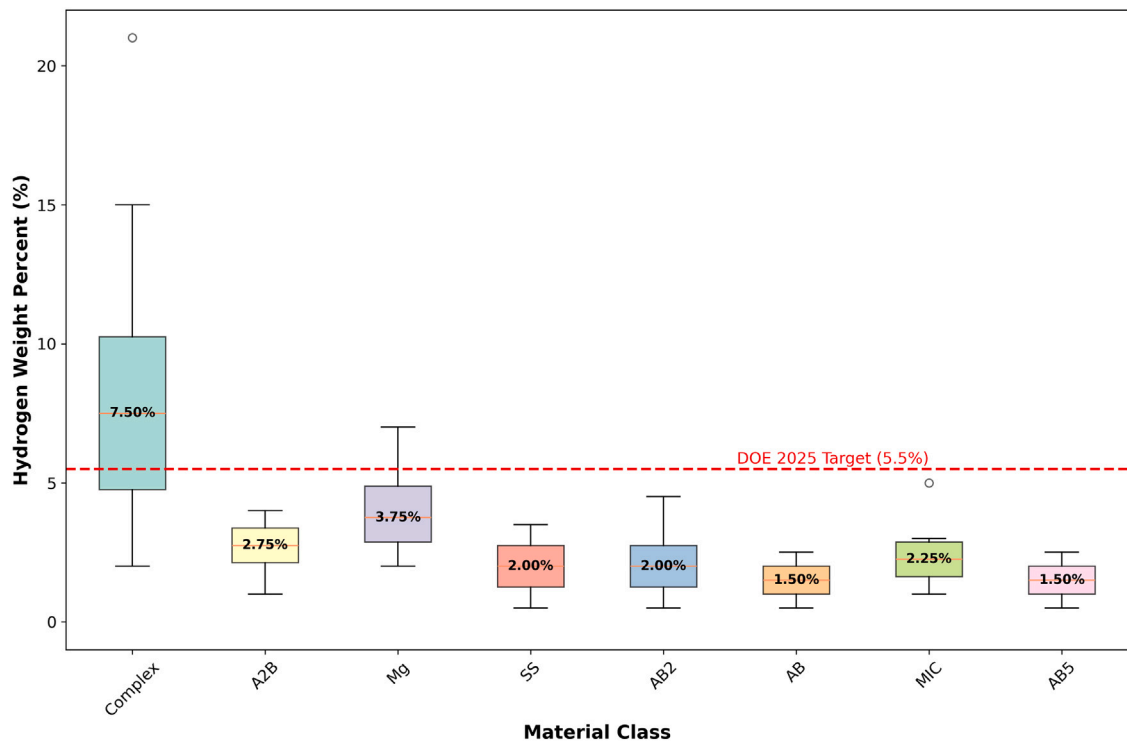


Fig. 11. Box plot comparing hydrogen weight percent across different material classes. Each box represents the interquartile range (IQR), the bold black line in the box marks the median, and whiskers extend to capture non-outlier values. Outliers appear as individual points. The black outlines accentuate the box boundaries, while each material class is distinguished by a color from the 'viridis' palette.

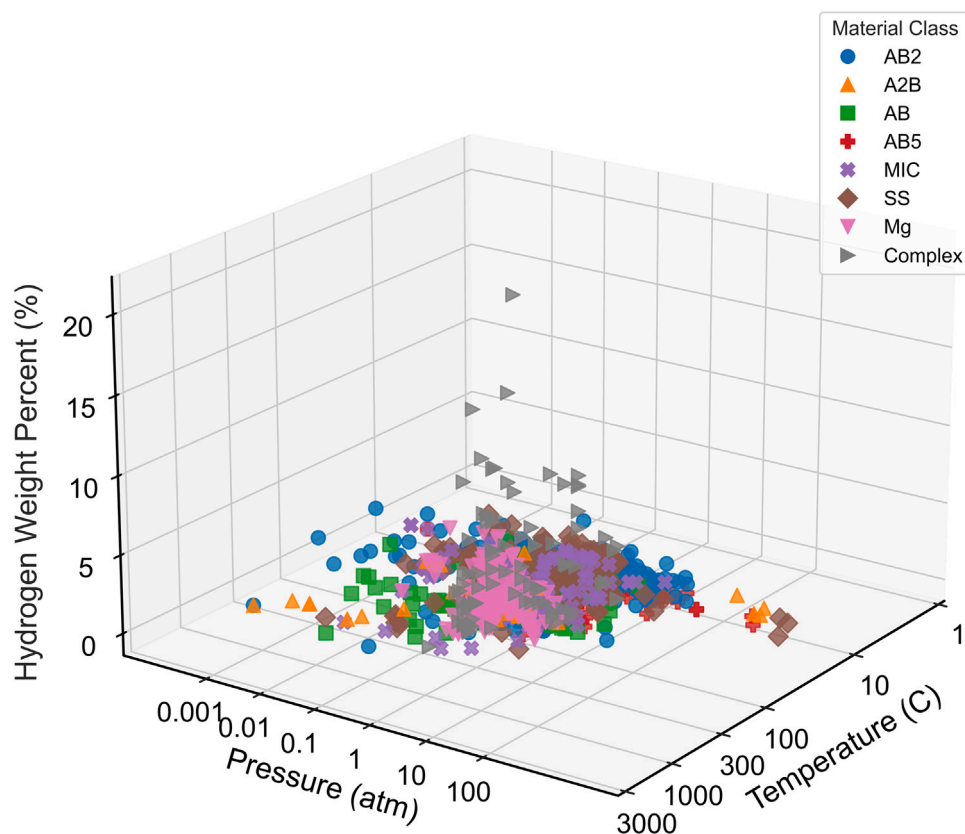


Fig. 12. 3D scatter plot of hydrogen weight percent as a function of temperature and pressure, both represented on log scales. Each material class is indicated by a unique marker shape and color. Notably, among the various classes, Complex hydrides exhibit comparatively higher storage capacities under moderate temperature and pressure conditions.

enthalpy requires significant heat input for hydrogen release, posing a challenge for practical deployment.

Toluene/methylcyclohexane (MCH) is another widely researched LOHC system, where toluene is hydrogenated to MCH, storing approximately 6.1 wt% hydrogen. The reversible cycle between toluene and MCH has been utilized in laboratory and industrial applications due to its well-established production and handling infrastructure [93]. However, dehydrogenation of MCH requires temperatures around 300 °C and the use of noble metal catalysts, typically Pt- or Pd-based. More recently, dibenzyltoluene (DBT) and its derivatives have gained attention for their suitability as LOHCs, offering a hydrogen capacity of ~ 6.2 wt%. Ojelade & Zaman [145] conducted a comprehensive thermodynamic analysis of the methylcyclohexane–toluene system, demonstrating almost 100% equilibrium conversion of methylcyclohexane with ~99% H₂ yield at 325 °C and 1 bar during dehydrogenation. For the reverse toluene hydrogenation, calculations predict near-complete conversion at 100 °C and 1 bar, though kinetic limitations in current catalysts prevent realizing these theoretical values at atmospheric pressure. These compounds remain liquid in both hydrogen-lean and hydrogen-rich states, possess low vapor pressure—which enhances safety by reducing flammability risks—and exhibit strong thermal stability, making them viable for large-scale applications [146].

LOHCs present several advantages in hydrogen storage and transport. They achieve high volumetric hydrogen densities comparable to liquid hydrogen but without requiring cryogenic temperatures. Because LOHCs remain in liquid form under ambient conditions, they can be integrated into existing fuel storage and distribution infrastructure, reducing the need for new systems. In their hydrogen-lean form, most LOHCs are non-flammable, non-explosive, and exhibit low vapor pressure, minimizing safety concerns. Additionally, the reversibility of the hydrogenation/dehydrogenation cycle allows LOHCs to be reused multiple times, unlike one-time-use chemical hydrides [142].

Despite these advantages, several challenges must be addressed for widespread adoption of LOHC technology. A major drawback is the endothermic nature of dehydrogenation, which requires substantial heat input, typically in the range of 150–300 °C, depending on the carrier [141]. Efficient heat management and external heat sources—such as waste heat recovery—are necessary for continuous operation. Another limitation is the reliance on noble metal catalysts, such as Pd, Pt, and Ru, to facilitate hydrogenation and dehydrogenation reactions. These catalysts are expensive and susceptible to poisoning from sulfur and other impurities, necessitating high-purity LOHCs. Research into non-noble metal alternatives, such as Ni-based catalysts, is ongoing, though noble metals currently provide superior performance [147].

The hydrogen release rate and overall efficiency also face challenges, particularly for applications requiring rapid hydrogen supply, such as fuel cells. Reactor designs must optimize heat transfer and catalysis to ensure sufficient hydrogen release while maintaining energy efficiency. The logistics of LOHC recycling further complicate implementation, as the spent hydrogen-lean carrier must be transported back to hydrogenation facilities, effectively requiring a closed-loop supply chain [146]. This adds cost and complexity, as tanker trucks or pipelines would need to carry hydrogen-rich LOHCs to fueling stations and return with hydrogen-lean LOHCs for reprocessing. Additionally, partial dehydrogenation and by-product formation can lead to multiple hydrogenated species or intermediates, which complicate hydrogen utilization and necessitate precise control of reaction conditions.

Ongoing research focuses on addressing these challenges through improved catalysts and the discovery of new LOHC molecules with optimized properties. Developing catalysts with higher activity and selectivity can lower the temperature required for dehydrogenation while increasing hydrogen release rates. Exploring cheaper metal alternatives or structured bimetallic catalysts may also reduce costs [146]. Another promising avenue is the design of LOHC molecules with slightly

lower hydrogen capacities but significantly lower dehydrogenation enthalpies, which could improve efficiency. Recent studies have investigated heterocycles, unsaturated nitrogen compounds, and eutectic mixtures of LOHCs to enhance performance [141]. Additionally, integrating LOHC systems with renewable or industrial waste heat sources is being explored to improve overall energy efficiency. For instance, utilizing surplus heat from power plants to drive the hydrogen release process could significantly enhance the viability of LOHC-based hydrogen storage [148].

3.5.1. Formic acid as a hydrogen storage medium

Formic acid (HCOOH) functions as a specialized type of LOHC, storing hydrogen in a simple liquid that decomposes into hydrogen and carbon dioxide under relatively mild conditions [149]. Although its hydrogen content is only 4.4 wt%, which is modest compared to other hydrogen carriers, its primary advantage lies in the low-temperature release of hydrogen, typically between 70 and 120 °C with appropriate catalysts. Additionally, formic acid remains a liquid under ambient conditions, making it easy to handle and integrate into existing fuel infrastructure.

Catalytic decomposition of formic acid has been extensively studied, demonstrating efficient hydrogen release. For instance, Czaun et al. [150] generated the active catalyst in situ from $\text{RuCl}_3 \cdot x\text{H}_2\text{O}$ and triphenylphosphine in an SDS-stabilized toluene/water emulsion. At 90 °C they achieved complete conversion of formic acid to an equimolar H_2/CO_2 stream with no detectable CO; the initial rate ($1.69 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}$ with 2.43 mM Ru) corresponds to a turnover frequency of $\approx 25 \text{ h}^{-1}$, rising to $\approx 100 \text{ h}^{-1}$ at 100 °C.

A key challenge, however, is the production of CO_2 as a byproduct. If this CO_2 can be captured or chemically recycled—such as through hydrogenation back into formic acid using renewable hydrogen—the overall process could become carbon-neutral. Filonenko et al. [151] demonstrated a ruthenium-based catalyst capable of hydrogenating CO_2 to formic acid with a TON up to 1,100,000 and a TOF up to $23,000 \text{ h}^{-1}$ at 120 °C and 40 bar ($\text{H}_2/\text{CO}_2 = 1:1$). However, the reaction has $\Delta G^\circ = +32.9 \text{ kJ mol}^{-1}$ at standard conditions, necessitating the use of a base or other techniques to shift equilibrium for practical application. Recent research has focused on improving both decomposition and synthesis catalysts, as well as investigating electrocatalytic pathways for directly converting CO_2 to formate using renewable electricity, potentially simplifying the hydrogen storage cycle and enhancing its environmental sustainability [152].

Recent advances in formic acid-based hydrogen storage have brought the technology closer to practical application. Sang et al. (2024) introduced a potassium formate/potassium bicarbonate system that remained stable over six months and completed 40 full storage-release cycles [153]. Using only trace amounts of ruthenium catalyst, the system released hydrogen at a rate of 9.3 L/h with 99.5% purity and achieved a potential storage efficiency of up to 67%. Operated under mild conditions without additional chemical inputs, this carbon-neutral approach stores hydrogen in benign salt solutions. Photocatalytic methods have also advanced significantly, with Peng et al. (2024) reporting light-driven dehydrogenation of formic acid at activity levels 100 times higher than previous photocatalytic systems [154]. Their system operates at room temperature using sunlight, eliminating the need for external heating. Notably, they demonstrated the coupling of photocatalytic hydrogen generation with enzymatic synthesis to produce high-value chemical and pharmaceutical compounds, marking the first such integration of renewable hydrogen production with sustainable chemical manufacturing. Ongoing efforts continue to address limitations by developing more efficient catalysts, refining decomposition mechanisms, and exploring advanced CO_2 capture and recycling strategies. These improvements are essential for transitioning formic acid-based systems from a promising concept to an economically viable component of hydrogen energy infrastructure. More broadly, LOHCs remain a key research focus, with future directions aimed at reducing catalyst costs, optimizing carrier molecules, and integrating these systems with renewable energy and waste heat recovery to improve efficiency and overall sustainability [155].

3.6. Photocatalytic and light-driven storage systems

Light-driven hydrogen storage systems offer the potential to directly harness solar energy for hydrogen production and storage processes, enabling fully sustainable energy cycles. Recent advances have shown that photocatalytic materials can effectively drive hydrogen storage and release reactions using only sunlight, removing the need for external energy inputs [156]. These systems typically rely on advanced photocatalysts that convert light energy directly into chemical energy for hydrogen liberation, with some materials now capable of utilizing near-infrared light to extend the usable solar spectrum beyond traditional visible-light systems. The integration of photocatalytic processes with hydrogen storage materials has opened new possibilities for distributed energy systems and off-grid applications. Advanced photocatalysts demonstrate improved performance under real solar conditions and hold promise for practical outdoor deployment [156]. The development of hybrid photocatalytic-storage systems allows continuous operation during daylight hours while storing hydrogen for later use, creating a complete solar-to-hydrogen solution. These technologies are especially valuable for remote or developing regions where access to conventional energy infrastructure is limited.

4. Summary of hydrogen storage methods

Table 7 presents a comparison of established and emerging hydrogen storage technologies, including those discussed earlier. The aim is to highlight each technology's principal advantages and disadvantages, typical operating conditions, achievable gravimetric storage capacities, and current development stage. From compressed and liquid hydrogen to advanced nanostructured materials and chemical carriers, this overview provides a clear snapshot of the current research landscape and practical applications in hydrogen storage.

Fig. 13 provides a comprehensive comparison of hydrogen storage performance across all major material classes relative to established DOE targets. Complex metal hydrides demonstrate the highest maximum storage capacity ($> 20 \text{ wt}\%$), significantly exceeding the DOE Ultimate target of 6.5%, while several physical storage materials including PCN and activated carbon approach or exceed the DOE 2025 target of 5.5%. The analysis reveals that chemical storage methods (orange bars) generally achieve higher maximum capacities but with greater variability, whereas physical storage methods (blue bars) show more consistent but lower performance levels. Notably, most material classes fall short of the DOE 2020 target of 4.5% in their mean performance, highlighting the ongoing challenges in developing practical hydrogen storage systems that meet commercial viability requirements.

5. Techno-economic analysis

The economic viability of hydrogen-storage technologies represents a critical factor determining their commercial adoption alongside technical performance metrics. While the techno-economic analysis presented in 8 provides system-level cost comparisons, a brief examination of the underlying economic factors reveals important considerations for technology development and deployment.

Cost analysis reveals significant variations across storage technologies, primarily driven by material requirements and system complexity. Compressed-gas systems exhibit moderate capital costs of approximately 500–700 USD per kilogram of hydrogen capacity, dominated by high-pressure composite vessels that require expensive carbon-fiber materials [163]. Metal hydrides present a wide cost spectrum, with simple magnesium-based systems offering low material costs (10–30 USD per kilogram) but requiring substantial thermal-management infrastructure, while advanced rare-earth alloys can cost 100–1000 USD per kilogram of material [164]. Emerging technologies such as metal-organic frameworks face high synthesis costs due to expensive organic linkers and specialized processing requirements, though these costs

Table 7

Comprehensive comparison of hydrogen storage methods.

Storage method	Description	Advantages	Disadvantages	Operational conditions	Hydrogen storage capacity	Development stage
Compressed hydrogen storage	Hydrogen gas stored at high pressures (up to 700 bars) in specially designed tanks made of metals or composite materials. Types include Type I (metal), Type II (metal with composite overwrap), Type III (composite with metal liner), and Type IV (all composite).	- Simple and established technology - Immediate applicability - High filling and extraction speeds - Well-established infrastructure	- Requires robust, heavy tanks - Less space-efficient - High cost of compression - Safety concerns with high-pressure hydrogen gas - High energy consumption for compression	- Pressure: Up to 700 bar - Temperature: 15–25 °C (288–298 K)	- Gravimetric: 1–3 wt% - Volumetric: Up to 36 kg·m⁻³	Commercial use; ongoing research into lightweight composite materials
Liquid hydrogen storage	Hydrogen is liquefied at cryogenic temperatures (−253°C) and stored in thermally insulated cryogenic tanks designed to minimize heat transfer and prevent boil-off.	- Higher energy density by volume compared to compressed gas - High volumetric hydrogen density (70.8 kg·m⁻³) - Suitable for space-constrained applications	- Very low temperatures required, resulting in high energy consumption for liquefaction (~30% of hydrogen's energy content) - Potential boil-off losses - Requires complex insulated storage tanks - Safety concerns with handling liquid hydrogen	- Pressure: 1–5 bar (0.1–0.5 MPa) - Temperature: −253°C (20 K)	- High volumetric capacity - Gravimetric capacity affected by tank mass and insulation	Used in specialized applications; limited commercial use due to cost and technical challenges
Physisorption in Porous Materials	a. Carbon-Based Nanostructures: Hydrogen adsorbed onto materials like activated carbons, CNTs, and graphene via van der Waals forces. b. Metal-Organic Frameworks (MOFs): Crystalline materials with exceptionally high surface areas and tunable pore sizes. c. Covalent Organic Frameworks (COFs): Porous organic materials with high surface areas and low densities.	- High surface area materials - Lightweight and chemically stable - Tunable structures - Safer, lower-pressure storage - Potential for modular design	- Require cryogenic temperatures (77 K) for significant uptake - Low uptake at 15–25 °C (288–298 K) - Lower storage capacities compared to other methods - Sensitive to temperature and pressure changes - Some materials may be expensive or unstable	- Temperature: 77 K - Pressure: Up to 10 MPa (varies with material)	- Carbon materials: Up to 7 wt% at 77 K - MOFs: Up to 9 wt% at 77 K - COFs: Up to 7.2 wt% at 77 K	Research/Lab scale
Metal hydrides	Hydrogen stored by forming solid compounds where hydrogen is absorbed into the metal lattice; examples include magnesium hydride (MgH ₂), sodium alanate (NaAlH ₄), and lithium hydride (LiH).	- High volumetric hydrogen density (up to 115 kg·m⁻³) - Safe and compact storage - Reversible hydrogen absorption and desorption under moderate conditions	- Low gravimetric capacity (<3–8 wt%) due to heavy metals - Heavy materials limit use in mobile applications - Slow kinetics of hydrogen absorption/desorption - High desorption temperatures - Heat management required	- Pressure: 1–5 bar (0.1–0.5 MPa) - Temperature: 15–25 °C (288–298 K) to 300–400 °C (573–673 K)	- Gravimetric: Up to 7.6 wt% (MgH₂ theoretical) - Volumetric: Up to 115 kg·m⁻³	Commercial use in niche applications; ongoing research to improve materials and kinetics
Complex hydrides	Chemical compounds that store hydrogen in complex anions; examples include lithium borohydride (LiBH ₄), sodium borohydride (NaBH ₄), and sodium alanate (NaAlH ₄).	- High gravimetric hydrogen capacity (up to 18 wt%) - Potential for reversible storage with catalysts (e.g., Ti-doped NaAlH₄) - Low cost	- High desorption temperatures (> 250 °C) - Slow kinetics of hydrogen release - Irreversibility and degradation over cycles - Challenges in rehydrogenation processes	- Pressure: 5–50 bar (0.5–5 MPa) - Temperature: High (> 250 °C)	- Gravimetric: Up to 18 wt% (LiBH₄) - N-ethylcarbazole: 5.8 wt% theoretical	Research stage; significant challenges remain for practical application

(continued on next page)

may decrease with scale-up and process optimization [47]. Liquid-organic hydrogen carriers benefit from relatively low carrier-fluid costs (3–8 USD per kilogram) but require capital-intensive hydrogenation and dehydrogenation facilities, while ammonia systems leverage

existing production infrastructure despite energy-intensive synthesis and cracking processes [146].

Manufacturing complexities vary widely across technologies and significantly impact scalability. Compressed-gas storage relies on

Table 7 (continued).

Storage method	Description	Advantages	Disadvantages	Operational conditions	Hydrogen storage capacity	Development stage
Liquid Organic Hydrogen Carriers (LOHCs)	Organic liquids that can absorb and release hydrogen through reversible hydrogenation and dehydrogenation reactions; examples include N-ethylcarbazole and toluene/methylcyclohexane systems.	• High storage capacity (5–7 wt%) • Liquid at 15–25 °C (288–298 K) (easy to transport) • Safe and convenient transport • Utilizes existing infrastructure • Reversible processes	• Requires catalysts (often expensive noble metals) • High dehydrogenation temperatures (180–300 °C) • Endothermic dehydrogenation reactions • Low efficiency of dehydrogenation • Potential environmental impacts	• Hydrogenation: – Pressure: Up to 6 MPa – Temperature: 180–200 °C • Dehydrogenation: – Pressure: Atmospheric – Temperature: 180–300 °C	• Toluene/methylcyclohexane: ~6.2 wt%	Research stage; ongoing efforts to improve catalysts and processes
Nanostructured Materials	Utilizing nanostructured materials (e.g., MgH ₂ nanoparticles, nanowires) to enhance hydrogen storage properties through increased surface area and altered thermodynamics.	• Improved hydrogen absorption/desorption kinetics • Lower dehydrogenation temperatures • Enhanced surface reactivity	• Potential for particle agglomeration • Complex synthesis methods • Stability issues over cycles • Challenges in scalability	• Dehydrogenation starting at ~ 85 °C (Mg nanoparticles) • Pressure: 2–10 MPa (20–100 bar)	• MgH ₂ theoretical capacity: 7.6 wt%	Research/Lab scale
Glass Microspheres Storage	Hydrogen stored in small, hollow glass microspheres; hydrogen diffuses into the microspheres at high temperatures and pressures and is released upon heating.	• High hydrogen capacity (> 15 wt%) • Good maneuverability	• Immature technology • Challenges in producing high-strength microspheres • High cost • Requires >100 °C (>373 K) for loading/unloading	• Pressure: >50 bar (>5 MPa) internal pressure • Temperature: >100 °C (>373 K) for loading/unloading	• Gravimetric: > 15 wt%	Research stage; not yet practical for widespread use

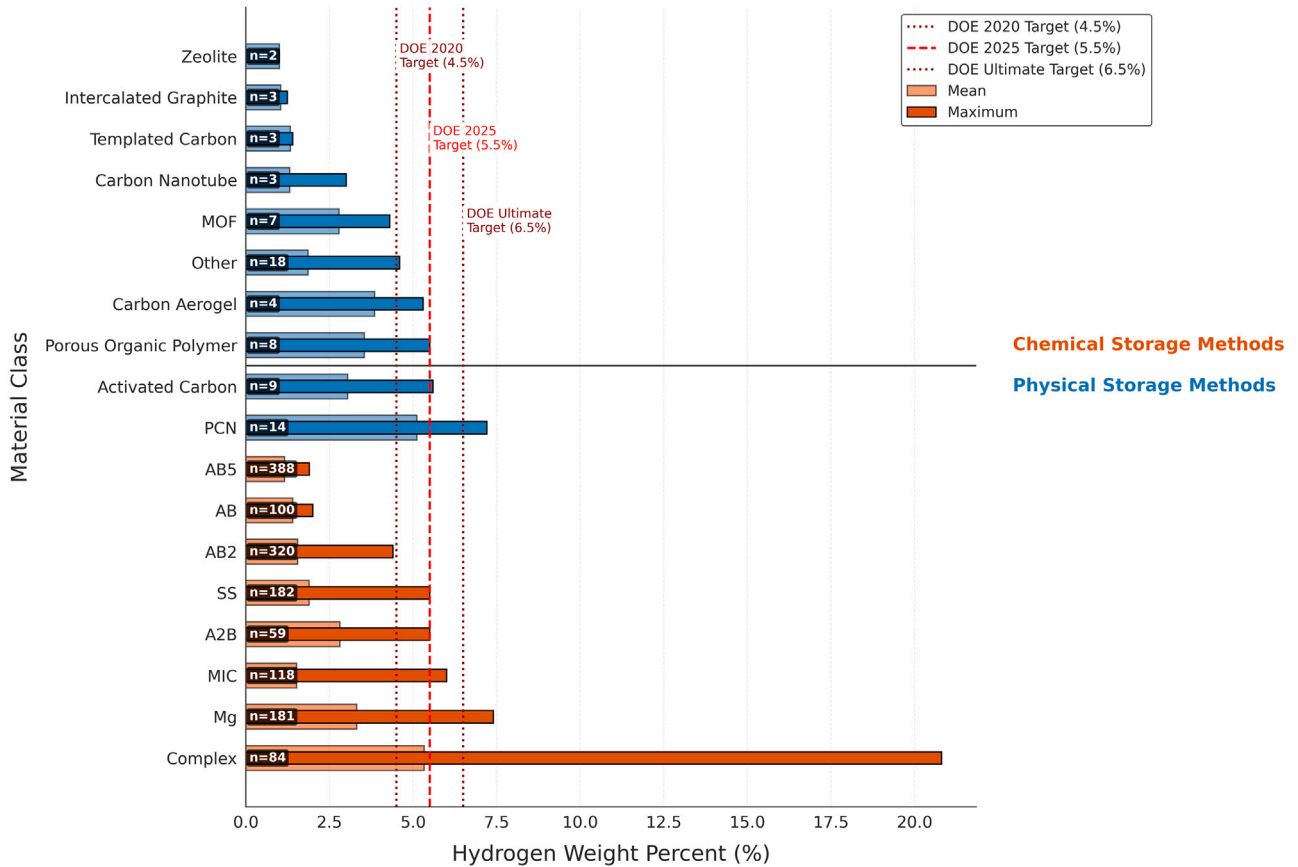


Fig. 13. Comparison of hydrogen storage performance across material classes relative to DOE targets. Horizontal bars show mean (light) and maximum (dark) hydrogen weight percentages for physical storage methods (blue) and chemical storage methods (orange). Vertical dashed lines indicate DOE 2020 (4.5%), 2025 (5.5%), and Ultimate (6.5%) targets. Sample sizes (*n*) are indicated for each material class. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 8
Comparison of hydrogen storage methods.

Storage method	Approx. system cost	Energy efficiency (round-trip)	Safety & Infrastructure	References
Compressed gas	\$500–700/kg H ₂ capacity (\$15–20/kWh); high-pressure compressor needed	~80%–90% (compression uses 5%–20% of energy)	Proven and modular; high-pressure hazards (mitigated by tank design); requires robust cylinders and 700 bar fueling stations.	[157,158]
Liquid H ₂	\$30–50/kg H ₂ capacity (tank) + ~\$1–2/kg liquefaction	~60%–70% (liquefaction consumes 30%–40%)	High density; boil-off requires venting or re-liquefaction; cryogenic safety protocols needed; existing LH ₂ transport infrastructure limited to industrial use.	[159,160]
Cryo-Compressed H ₂	~\$300–400/kg H ₂ (projected)	~70%–80% (liquefaction to ~80 K, minimal boil-off losses)	Promising hybrid: extended dormancy with no venting (Rivard et al. 2019); requires cryogenic fueling; combines some complexity of both pressurized and cryogenic systems.	[161]
Metal Hydrides	Highly variable (material-dependent; e.g. \$10–30/kg for MgH ₂ material) + heat exchangers	~70%–90% (depends on heat management; nearly isothermal storage)	Low pressure operation (safer); heavy systems; needs thermal input for H ₂ release; best for stationary or vehicles with waste heat.	[157,162]
Adsorbents (MOF/Carbon)	Lab-scale costs high (MOFs can be costly; activated carbon cheaper) – not yet optimized for cost	~70%–85% with cryo-cooling (e.g. LN ₂ cooling overhead)	Low/moderate pressure; requires cooling to ~77 K for good capacity; safety similar to compressed gas but lower pressure; no commercial systems yet.	[160]
LOHC (e.g. DBT)	Hydrogenation/dehydrogenation units: large capital; LOHC fluid ~\$5/kg (reused)	~60%–70% (35%–40% energy loss as heat)	Ambient liquid – easy to handle with oil infrastructure; no H ₂ loss over time; requires chemical plant and catalysts at both ends. Not suitable for onboard/mobile release.	[142,160]
Ammonia	Haber-Bosch plants (capital-intensive); NH ₃ transport cheap (existing); NH ₃ cracking unit adds cost	~50%–60% (30% loss in synth + 10%–20% in cracking)	Widely handled chemical (toxic, requires training); very high volumetric density; can use existing tanks/pipelines; versatile end-use (crack to H ₂ or direct fuel).	[162]

established composite-manufacturing techniques but requires precise filament-winding processes and specialized carbon-fiber materials with limited global production capacity. Metal-hydride production involves controlled-atmosphere processing to prevent oxidation and specialized heat-treatment procedures, with manufacturing costs increasing substantially for complex alloy compositions. Advanced adsorbent materials face challenges in maintaining consistent properties during scale-up, as laboratory synthesis methods often fail to translate effectively to industrial production. Chemical storage systems require high-pressure, high-temperature processing capabilities and advanced catalyst systems, adding complexity to manufacturing operations [165].

Scalability considerations encompass both technical and economic factors that influence the transition from laboratory development to commercial deployment. Learning-curve effects observed in energy-storage technologies suggest potential for significant cost reductions through increased production volumes, with typical cost reductions of 15%–25% per doubling of cumulative production [166]. However, realizing these benefits requires sustained market development and substantial capital investment for manufacturing scale-up. Supply-chain constraints present additional challenges, particularly for technologies dependent on rare materials or specialized components with limited production capacity. The geographic concentration of key materials—such as rare-earth elements for advanced metal hydrides or carbon fiber for pressure vessels—creates potential supply vulnerabilities that must be considered in long-term economic planning.

The integration of hydrogen storage systems with renewable energy sources presents both opportunities and challenges for economic optimization. Variable renewable-electricity costs can improve the economics of energy-intensive storage methods during periods of excess generation, while the need for flexible operation across varying charging schedules may complicate system design and increase costs. Market mechanisms, including time-of-use pricing and carbon-pricing policies, significantly influence the relative economic attractiveness of different storage technologies and their competitive position relative to alternative energy-storage and transportation-fuel options [167].

The various hydrogen storage methods outlined above each occupy different “sweet spots” in terms of performance and cost. Table 8 summarizes several key techno-economic metrics – approximate ranges for system cost, energy efficiency, and notable safety/infrastructure notes – to facilitate comparison. While exact numbers vary with system design and scale, general trends can be observed:

Table 8 highlights the trade-offs across various hydrogen storage technologies, illustrating how system cost, energy efficiency, and safety or infrastructure requirements differ significantly between approaches. Compressed gas systems are mature and modular, with moderate capital costs (approximately \$500–700 per kg H₂ capacity), but they rely on robust high-pressure vessels and dedicated fueling infrastructure. Liquid hydrogen offers high energy density, yet the liquefaction process imposes a significant energy penalty of about 30%–40%. Cryo-compressed hydrogen represents a hybrid solution that may reduce

boil-off losses, although it inherits operational challenges from both high-pressure and cryogenic systems.

Metal hydrides show a broad cost spectrum depending on the material—such as Mg-based alloys—and require effective thermal management to enable hydrogen release. Adsorbent-based systems, particularly those using advanced metal–organic frameworks (MOFs), remain in the research phase and face hurdles related to cryogenic operation and infrastructure scalability. LOHC technologies benefit from compatibility with existing petrochemical infrastructure, but the hydrogenation and dehydrogenation steps are energy-intensive, limiting their viability in mobile platforms. Ammonia-based storage stands out for its established production and distribution network; however, the toxicity and safety concerns associated with NH_3 , along with the energy losses incurred during synthesis and cracking (around 40%–50%), remain important considerations.

6. Life cycle assessment of hydrogen storage technologies

Life cycle assessment (LCA) quantifies the environmental burdens of hydrogen storage technologies across material extraction, production, operation, and end-of-life stages [2]. Key environmental impacts identified include energy consumption, greenhouse gas emissions, resource depletion, and potential pollution.

Compressed and liquid hydrogen storage methods require substantial energy for compression and liquefaction, which lead to significant carbon footprints, particularly when electricity is sourced from fossil fuels [5]. Moreover, the manufacturing of storage tanks and cryogenic equipment contributes to environmental impacts due to material production and processing. Chemical storage methods, such as the production of metal hydrides and complex hydrides, are also energy-intensive and may involve hazardous chemicals, resulting in emissions and waste [168]. The recycling and disposal of spent materials from these processes pose additional environmental challenges. Advanced nanostructured materials require specialized synthesis techniques and chemicals, which contribute to energy consumption and potential environmental impacts; however, the improved efficiency and performance of these materials may offset these impacts over their life cycle [169].

To mitigate the environmental impacts of hydrogen storage technologies, a multifaceted approach involving technological advancements, sustainable practices, and policy interventions is essential. Improving energy efficiency is paramount; developing more efficient storage technologies and processes can significantly reduce energy consumption. This includes innovations in compression technologies, liquefaction processes, and material synthesis to enhance overall efficiency [33]. Utilizing renewable energy sources for hydrogen production, compression, liquefaction, and regeneration processes substantially lowers the carbon footprint as integration with renewable energy systems enhances sustainability [170]. Establishing comprehensive recycling systems for storage equipment materials is also crucial in reducing waste and environmental pollution. Advances in recycling technologies for composite materials, metals, and nanomaterials are essential to facilitate this process [6]. Adopting environmentally friendly manufacturing practices, such as green chemistry principles and clean production technologies, minimizes harmful emissions, making the development of scalable, low-impact synthesis methods for storage materials critical [21].

Implementing strict environmental policies and industry standards helps regulate production, implementation, and waste disposal processes, while incentivizing the adoption of green technologies and sustainable practices promotes industry-wide improvements [2]. Enhancing safety protocols and implementing advanced monitoring technologies are vital to prevent leaks and environmental contamination; in underground storage scenarios, real-time monitoring of pressure, gas composition, and subsurface conditions is imperative [171]. Finally, investing in research and development to create new materials with higher capacities, better kinetics, and lower environmental impacts

is essential. Multidisciplinary approaches that combine materials science, engineering, and environmental science can drive innovation and contribute significantly to reducing the environmental footprint of hydrogen storage.

The quantitative comparison presented in Table 9 demonstrates the substantial range of environmental impacts associated with different hydrogen storage approaches. The data reveal that material production and manufacturing processes dominate the life cycle carbon footprint for most technologies, with operational emissions typically representing a smaller fraction of the total environmental burden. This finding has important implications for technology selection and development priorities, suggesting that efforts to reduce manufacturing emissions may yield greater environmental benefits than focusing solely on operational efficiency improvements.

The pressure dependency observed in compressed gas storage systems illustrates the environmental trade-offs associated with achieving higher storage densities. While 700 bar systems offer improved volumetric storage capacity compared to 350 bar alternatives, this advantage comes at the cost of a 65% increase in life cycle carbon emissions, highlighting the importance of optimizing storage pressure based on specific application requirements and environmental constraints rather than simply maximizing storage density.

The exceptionally high carbon footprint of metal hydride storage systems—nearly an order of magnitude greater than compressed gas alternatives—raises important questions about the environmental viability of this technology for large-scale deployment. While metal hydrides offer operational advantages such as low-pressure operation and inherent safety characteristics, these benefits must be weighed against the substantial environmental costs associated with material production and system manufacturing.

The moderate performance of LOHC and ammonia storage systems makes them promising alternatives for applications where the environmental impact of compressed gas storage is acceptable, but the operational demands of liquid hydrogen are impractical. The established industrial infrastructure for ammonia production and handling may provide opportunities to leverage existing supply chains and manufacturing capabilities, potentially reducing environmental impact through economies of scale and process optimization.

Regional manufacturing considerations are a critical factor in performance optimization, particularly for energy-intensive production processes. The wide variation in carbon fiber production emissions based on manufacturing location suggests that strategic placement of production facilities in regions with low-carbon electricity grids could significantly reduce the environmental impact of compressed gas storage systems; similarly, production of metal hydride alloys and MOF materials could benefit from regions with abundant renewable energy resources.

Finally, a life cycle perspective reveals opportunities for environmental impact reduction through improved material utilization and system design optimization. Extended service lives, enhanced recyclability, and improved manufacturing efficiency all represent pathways to reduce the per-unit environmental impact of hydrogen storage technologies. Development of circular economy approaches for storage materials—particularly for high-impact technologies such as metal hydrides and MOFs—could substantially improve their environmental performance over multiple use cycles.

7. Safety considerations in hydrogen storage technologies

Hydrogen leakage is a persistent challenge due to hydrogen's small molecular size and lack of sensory indicators. Even well-engineered systems can leak through microscopic openings [178]. While minor leaks may not create immediate hazards, accumulation in enclosed spaces can lead to flammable mixtures or, in rare cases, asphyxiation [179]. Unlike natural gas, hydrogen cannot be effectively odorized for safety because common additives may separate or interfere with

Table 9
Life cycle carbon footprint comparison of hydrogen storage technologies.

Storage technology	Carbon footprint (kg CO ₂ -eq/kg H ₂)	Primary contributing factors	Environmental sustainability	References
Compressed gas (350 bar)	200	Aluminum extraction, carbon fiber production	Most sustainable	[172,173]
Compressed gas (700 bar)	329	Higher carbon fiber content, pressure vessel materials	Moderate	[172,173]
Liquid hydrogen	150–250	Energy-intensive liquefaction, cryogenic infrastructure	Moderate	[174,175]
LOHC Storage	300–500	Organic carrier synthesis, catalyst production	Moderate	[174,176]
Ammonia storage	450–650	Haber–Bosch process, high-pressure infrastructure	Moderate–high	[174,177]
MOF-based storage	800–1200	Complex synthesis, organic ligand production	High impact	[172,174]
Metal hydride storage	2057	Energy-intensive alloy production, steel vessel	Highest impact	[172,173]
Gasoline tank (steel)	3.5	Steel production and forming	Reference comparison	[173]
Gasoline tank (plastic)	1.8	HDPE production and forming	Reference comparison	[173]

fuel cells and catalysts [13]. In compressed gas storage (350–700 bar), most leaks occur at joints and valves, not the tanks themselves [27]. Cryogenic systems face additional issues from material stress due to extreme cold. Leaks in liquid hydrogen are often spotted by white vapor clouds formed from condensed moisture, which can drift unpredictably before hydrogen warms and rises [13]. In solid-state systems like metal hydrides, MOFs, or LOHCs, hydrogen is chemically or physically bound, reducing the risk of leak during storage. However, leaks can still occur during charging or release. Impurities such as oxygen or water can trigger unwanted hydrogen release in some hydrides [99]. In LOHCs, hydrogen remains secure in storage but must be carefully managed during release; catalyst degradation can cause incomplete conversion and system instability [146]. Preventing leaks requires tailored approaches, routine inspection, compatible materials, and containment layers for compressed and cryogenic systems, and strict control of temperature and contamination for material-based storage [28].

Emergency Protocols: Hydrogen's wide flammability range (4%–75% in air) and low ignition energy (≈ 0.02 mJ) [9] demand strict emergency protocols for all storage systems, including reliable detection, personnel training, clear response procedures, and coordination with local responders. Modern facilities use multiple detection tools such as catalytic bead sensors for low concentrations and thermal conductivity detectors for higher ones [180]. According to hydrogen safety panel guidance, sensors should be placed near valves, ceilings, and escape routes [181], linked to alarms and emergency shutdown systems. Regular drills, as recommended by NFPA [182], are essential for readiness. Human error accounts for about 60% of incidents [13], so staff must be prepared to recognize hydrogen hazards, operate equipment properly, and respond to leaks or fires. Given that hydrogen flames are nearly invisible and emit little heat, firefighting protocols advise letting controlled hydrogen fires burn unless the leak can be stopped to avoid explosion risk [183]. Adequate ventilation—at least six air changes per hour with sensor-triggered increases—is vital for indoor systems [184]. Post-incident analysis should include root-cause investigation and sharing of lessons learned, as recommended by DOE's Safety Panel [185]. For larger sites, joint planning and training with first responders ensure effective external support and safe coordination during emergencies.

System Failures: Failures in hydrogen storage systems vary by technology but often result from mechanical stress, thermal imbalance, control faults, or material degradation. Compressed gas tanks can fail due to fatigue, impact damage, or design flaws. Though composite tanks undergo rigorous testing [186], analysis of 71 incidents revealed that over-pressurization, poor maintenance, or incompatible materials were often to blame [187]. Cryogenic systems are vulnerable to insulation loss, leading to rapid boil-off and pressure spikes. In hydride-based

systems, poor heat transfer during charging or discharging can reduce performance or damage materials [99]. Effective thermal design and monitoring are critical to avoid such failures.

Modern systems rely on sensors, valves, and logic controls to manage operation, but these can fail. One recorded incident involved premature rupture disc activation due to control malfunction, which safely vented the tank but emphasized the need for redundant safeguards [188]. Material embrittlement is another concern—hydrogen diffusion weakens some metals, cutting fracture toughness by up to 90% [189]. Proper material selection and inspection are essential. Impurities such as CO or moisture can also damage storage media or poison catalysts [146]. Mitigation includes redundant pressure relief devices, fail-safe system design, real-time monitoring, and strict adherence to safety standards like ISO 19881 and ASME Section VIII [186, 190]. A robust safety framework must combine technical safeguards with training, oversight, and a strong safety culture.

8. Conclusion, research gaps, and future directions

Hydrogen storage technologies have advanced considerably in recent years, with significant progress across diverse material systems and storage approaches. These advancements target longstanding challenges in efficiency, safety, and scalability, contributing to the broader shift toward a reliable and sustainable hydrogen economy. This comprehensive study of hydrogen storage technologies reveals that while significant progress has been made across multiple storage approaches, each method presents distinct trade-offs that affects its suitability for specific applications. Compressed gas storage offers technological maturity but requires high-pressure infrastructure, liquid hydrogen provides high density but demands continuous energy input, physical adsorption methods show promise but typically need low temperatures, and chemical storage approaches offer safety advantages but often require elevated temperatures for hydrogen release. Recent breakthroughs in formic acid systems, solid-state reticular materials, and photocatalytic approaches that harness renewable energy demonstrate significant advancements. However, the fundamental challenge of developing storage technologies that operate efficiently under ambient conditions is still persist.

The most critical research gap identified is the lack of ambient-condition hydrogen storage systems that achieve practical densities without extreme operational requirements, particularly constraining transportation and portable applications where normal operating conditions are essential. Immediate research priorities must focus on developing advanced materials with tailored thermodynamic properties

for room-temperature operation, integrating nanostructured systems to improve kinetics and reduce activation barriers, and creating hybrid storage systems that combine multiple mechanisms to optimize performance across different conditions. These efforts should prioritize novel metal hydride compositions with reduced enthalpy requirements, porous materials with enhanced ambient-temperature capacities, and cost-effective catalysts that enable efficient operation at moderate temperatures and pressures.

Advancing hydrogen storage technologies will require a strategic focus on several critical fronts: the development of materials that perform reliably under ambient conditions, the design of efficient catalysts for chemical storage systems, and the integration of lab-scale innovations into robust, deployable systems. Equally important are rigorous techno-economic analyses that assess large scale feasibility and guide investment decisions. Achieving these goals will depend on close collaboration among universities driving fundamental research, industry partners translating discoveries into engineering solutions, and government agencies providing targeted funding and regulatory support. Artificial intelligence and machine learning can transform hydrogen storage research by accelerating discovery and optimization across all areas. These methods accelerate the discovery process by enabling high-throughput screening of materials and identifying optimal operating parameters without the need for extensive experimentation. At the same time, hybrid storage systems that combine multiple mechanisms—such as physical and chemical storage, or solid-state and liquid-phase systems—are gaining attention for their ability to meet a wide range of application demands. These advances point toward a more flexible and application-specific future, where hydrogen storage technologies can be tailored to meet both performance requirements and operational constraints.

CRedit authorship contribution statement

Aimen Laalam: Writing – review & editing, Writing – original draft, Validation, Data curation, Conceptualization. **Parisa Bazazi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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